Probabilistic Narrative Identification: Evidence from Tax Policy*

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Preliminary and incomplete.
[Latest version here]

Abstract

This paper generalizes narrative identification by interpreting documentary classifications as probabilistic information about a latent exogeneity status. Traditional narrative approaches treat selected policy actions as fully exogenous, implicitly imposing a zero-contamination restriction on a chosen subset of observations. In contrast, the framework treats exogeneity as a latent binary attribute and interprets narrative evidence as probabilistic information about whether a given policy action reflects orthogonal variation. Identification exploits variation in narrative support to isolate the effects of policy changes that are most plausibly exogenous, without requiring the construction of observation-level structural shocks. The paper develops a document-grounded implementation that maps policymakers' stated motivations into a continuous index of narrative support for exogeneity while preserving traceability to underlying sources. Empirically, the paper validates this probability object in three steps. First, edited Romer–Romer narratives show that the procedure assigns higher support to episodes classified as exogenous in the canonical U.S. narrative series. Second, a full-pipeline UK exercise based on Cloyne's tax data shows that probabilities constructed from reconstructed policy documents align with an independent expert classification at the row level. Third, an end-to-end application to routine U.S. federal budget measures shows that document-based support can expand the Romer–Romer identification margin while producing contractionary tax-shock responses comparable to the benchmark series.

Keywords: Narrative identification; fiscal policy; tax shocks; external instruments; causal inference.
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Introduction

Evaluating the macroeconomic effects of fiscal policy is empirically challenging because policy actions typically respond endogenously to economic conditions. Legislated tax and spending changes reflect contemporaneous shocks, anticipated developments, and evolving policy objectives, confounding causal interpretation. A large literature has proposed identification strategies to address this problem, including structural VARs, external-instrument approaches, and narrative methods that exploit documentary evidence on policymakers' stated motivations. Within this landscape, narrative identification has played a central role in settings where credible instruments are scarce and policy intent must be inferred from historical records. Influential applications span fiscal and monetary policy and include work by Friedman and Schwartz (1963), Mertens and Ravn (2012), Ramey (2011), and Romer and Romer (1989, 2010, 2023). At the same time, narrative approaches remain contested: critics have questioned the credibility of sharp exogeneity classifications, the treatment of mixed motivations, and the sensitivity of results to discretionary inclusion decisions.

Traditional narrative identification maps documentary evidence into a binary inclusion rule. A subset of policy actions is retained when the historical record is judged to support an exogenous interpretation, and retained actions are then treated as fully insulated from endogenous responses. This approach has proved powerful in practice, but it embeds a knife-edge abstraction: once an action is classified as admissible, residual uncertainty about its motivations is ignored. In effect, selected actions are assumed to be free of endogenous contamination, while excluded actions are assumed to contain no usable identifying variation. In practice, however, fiscal measures frequently reflect mixed motivations, and the documentary record often provides only partial or ambiguous support for orthogonality. A binary rule therefore discards information about the intensity and heterogeneity of stated motives and concentrates identification on a small set of extreme cases.

This paper recasts narrative identification through a two latent-type representation that makes explicit the role of classification uncertainty. Rather than asking whether a policy action is exogenous or not, the framework interprets narrative evidence as shifting the probability that a given action belongs to an orthogonal latent class. Intermediate classifications therefore reflect uncertainty about latent exogeneity, not partial exogeneity. Identification proceeds by exploiting systematic variation in narrative support across policy actions. When documentary evidence is informative about how the likelihood of orthogonality varies across actions, the resulting cross-moment variation disciplines the causal interpretation of policy–outcome comovement, without requiring that any observed action be treated as perfectly exogenous. In particular, identification requires that stronger narrative support systematically increases the likelihood that a policy action belongs to the orthogonal latent class, while conditional on latent class membership, narrative support carries no additional information about contemporaneous outcome disturbances.

This perspective shifts the identifying content of narrative analysis away from a zero-contamination requirement for a selected subsample and toward moment restrictions involving imperfect information about latent orthogonality. Causal inference rests on how outcome–policy comovement changes with the strength of narrative support, not on the recovery of observation-level structural shocks. Section 1 formalizes this argument and characterizes the conditions under which variation in narrative support identifies the outcome–policy moment associated with the orthogonal latent class.

To operationalize this perspective, the paper develops a document-grounded implementation that translates qualitative historical evidence into an empirically usable measure of narrative support for exogeneity. The implementation separates the object being measured from the tax-policy protocol used to measure it. The object is a policy-level probability summarizing how the stated motivations in the contemporaneous documentary record support the orthogonal narrative class. The tax-policy implementation then specifies the semantic taxonomy, document-processing pipeline, calibration procedure, and aggregation rules used to construct that probability. The procedure preserves traceability to the underlying documentary record and incorporates diagnostics that clarify how the identifying variation used in estimation is supported by the available historical evidence.

The empirical analysis builds a validation ladder around the probability object. First, I apply the procedure to edited Romer–Romer narratives, removing the authors' explicit classification language and asking whether the remaining narrative evidence produces higher exogeneity support for episodes that Romer and Romer classify as exogenous. Second, I turn to Cloyne's UK tax dataset, where the full document-grounded pipeline can be evaluated against an independent expert classification at the policy-row level. This exercise tests whether probabilities generated from reconstructed UK policy documents align with Cloyne's endogenous and exogenous labels. Third, I apply the full procedure to a new dataset of routine U.S. federal tax measures drawn from the annual budget process, covering more than 1,000 policy actions outside the original Romer–Romer sample. The resulting budget-based shocks provide an end-to-end macroeconomic application and test whether documentary support can expand the support of a benchmark narrative shock series without overturning its central implications.

The paper makes three main contributions. First, it generalizes narrative identification by recasting narrative classifications as probabilistic information about latent exogeneity and by shifting identification toward a moment-based framework that does not require a perfectly exogenous subsample. Second, it develops a disciplined document-based measurement of narrative support that maps policymakers' stated motivations into an empirically usable index while preserving traceability to the underlying sources. Third, it provides a multi-benchmark validation of this probability object and an end-to-end U.S. application: the procedure aligns with expert narrative classifications in the Romer–Romer and Cloyne settings, and budget-based U.S. shocks constructed from the full document pipeline produce macroeconomic responses comparable to the canonical Romer–Romer benchmark.

Literature Review

Identification in Macroeconomics. The paper first situates itself within the literature on policy identification, which includes empirical strategies relying on reduced-form VARs identified through timing, sign, or other model-based restrictions (Blanchard & Perotti, 2002; Sims, 1980; Uhlig, 2005). A distinct line of work instead employs external instruments—variables correlated with the policy shock of interest but orthogonal to other structural innovations—to achieve identification (V. Ramey, 2016; Stock & Watson, 2018). Narrative approaches provide a primary methodology for constructing such instruments, using contemporaneous evidence on policymakers' stated motives. Foundational examples include Friedman and Schwartz (1963) for monetary policy and Hamilton (1985) for oil shocks, with subsequent applications (Ramey, 2011; Ramey & Shapiro, 1998; Ramey & Zubairy, 2018; Romer & Romer, 1989, 2004, 2010) establishing narrative identification as a leading empirical tool. A key limitation of these series is their extreme sparsity: as documented by Barnichon and Mesters (2025), typical narrative shock measures contain roughly 80 percent zeros, which substantially weakens statistical power. Their estimator proposes an econometric remedy for this power–credibility trade-off, but the underlying sparsity reflects, in part, the difficulty of constructing narrative shocks with limited coverage of smaller policy actions.

This paper contributes by reinterpreting narrative classifications as probabilistic information about exogeneity rather than as binary inclusion decisions. Traditional narrative approaches implicitly impose a zero-contamination restriction on a selected subset of policy actions. In contrast, the framework developed here treats exogeneity as a latent property and interprets documentary evidence as probabilistic information about the likelihood that a given action reflects orthogonal policy variation. Identification proceeds through observable cross-moments: variation in narrative support across policy actions isolates the outcome response associated with policy changes that are most plausibly exogenous, without requiring a perfectly exogenous subsample or the recovery of observation-level structural shocks. In this sense, the approach preserves the core logic of narrative identification, using historical evidence to assess orthogonality in expectation, while relaxing the knife-edge assumption that selected actions are free of endogenous contamination.

Text and AI in Economic Analysis. A growing literature uses text as an empirical input in economics, turning unstructured documents into structured variables such as beliefs, expectations, policy intent, or economic narratives. Foundational surveys emphasize that this requires defining the text unit, the target construct, the representation of text, and the validation design (Ash & Hansen, 2023; Gentzkow et al., 2019).

Recent LLM-based empirical work differs in how it treats the model output. One approach uses the LLM as a scalable research assistant, asking it to read documents and produce labels, summaries, rationales, or structured fields from text. This approach is useful when the goal is to reproduce or scale human coding of economic narratives (Clayton et al., 2025; Jamilov, 2025; Martell, 2025). A complementary approach treats the model as a source of statistical signals: classifier probabilities, token probabilities, embeddings, confidence measures, or model disagreement are retained and then calibrated, thresholded, or aggregated into economic variables (Buckmann & Hill, n.d.; Chen et al., 2026; Hartley, 2025).

This paper combines these two perspectives. The language model is used to organize contemporaneous fiscal narratives, but the final object is not a hard LLM label. Instead, model-generated textual signals are treated as noisy measurements of latent narrative exogeneity and are mapped into calibrated policy-level probabilities. This links the paper to recent concerns about validation and generated variables when AI-produced measurements enter econometric analysis (Battaglia et al., 2024; Carlson & Dell, 2025; Ludwig et al., 2025). Relative to existing LLM measurement papers, the contribution is to connect soft textual classification to narrative identification: the probability object is not only a confidence score or descriptive index, but an input into the moment restrictions that define the empirical object of interest.

Imperfect Moments and Measurement Robustness. The text and AI literature motivates why the constructed support index should be treated as a generated and potentially noisy variable. The moment-based formulation developed below determines how that noise enters the identification problem. Once narrative support enters the outcome–policy moment, residual classification error can be interpreted as a form of moment contamination. This connects the paper to econometric work on conditional moment restrictions (Chamberlain, 1987), plausibly exogenous or imperfect instruments

(Conley et al., 2012; Nevo & Rosen, 2012), corrected moments under measurement error (Evdokimov & Zeleneev, 2022), and nonclassical measurement error with multiple proxies (Hu & Schennach, 2008). Since LLM-generated variables may violate classical-error assumptions, the analysis also relates to approximate or misspecified moment conditions (Armstrong & Kolesár, 2021; Bonhomme & Weidner, 2022; Hansen & Lee, 2021). Relative to this literature, the contribution is to show how the credibility problem in narrative identification can be written as a moment-contamination problem, making these robustness tools directly relevant.

Fiscal Policy Transmission and Tax Composition. A final strand of literature concerns the macroeconomic transmission of fiscal policy. Seminal VAR-based analyses (Auerbach & Gorodnichenko, 2012; Blanchard & Perotti, 2002; Mountford & Uhlig, 2009) show that exogenous fiscal expansions raise activity, with multipliers varying across instruments and business-cycle conditions. Narrative studies of government spending shocks (Ramey, 2011; Ramey & Zubairy, 2018) highlight the role of expectations, while analogous work on taxation (Cloyne, 2013; Mertens & Ravn, 2012, 2013; Romer & Romer, 2010) shows that tax multipliers depend on timing, composition, and economic context. This literature also emphasizes that tax composition matters: income, corporate, payroll, and consumption-tax changes need not transmit through the same channels. That concern is central to the empirical validation below. The Cloyne exercise evaluates whether the probability pipeline recovers expert labels in a setting with rich tax-type heterogeneity, while the U.S. budget extension asks whether a dense set of routine federal tax measures can be filtered into macroeconomically meaningful identifying variation.

1 A Soft Model of Narrative Identification

The objective of this paper is to estimate the dynamic causal effects of legislated tax changes on macroeconomic outcomes using narrative evidence from policy documents. Let $\Delta\tau_t$ denote the aggregate legislated tax change in period t, and let y_{t+h} denote an outcome of interest at horizon $h \geq 0$. The structural relation of interest is

$$y_{t+h} = \beta_h \Delta\tau_t + \varepsilon_{t+h},$$

where β_h is the causal response of the outcome to an orthogonal tax innovation and ε_{t+h} collects other macroeconomic disturbances. Identification requires the relevant orthogonality condition, $\mathbb{E}(\Delta\tau_t\varepsilon_{t+h}) = 0$, to hold over the estimation sample; it fails when tax policy responds endogenously to current or anticipated conditions.

Traditional narrative identification addresses this problem by reading contemporaneous policy documents and classifying each tax change as either admissible or inadmissible for identification. That approach has proved influential in empirical work on tax policy, as in Romer and Romer (2010), Mertens and Ravn (2013), and Cloyne (2013), but it hardens a fundamentally uncertain historical judgment into a binary decision. The broader point is not specific to tax narratives. Other narrative fiscal instruments, such as defense-news shocks in Ramey and Zubairy (2018), are also motivated by historical judgments about plausibly exogenous sources of variation. Although major military buildups are often driven by geopolitical events rather than domestic stabilization policy, the resulting narrative signal need not be treated as exogenous with probability one: some defense-spending news may still be correlated with political, strategic, or macroeconomic conditions relevant for the outcome under study. Narrative evidence is therefore better viewed as providing probabilistic, rather than certain, identifying information.

This section takes that view literally. Narrative evidence is summarized by a continuous index $p_t \in [0,1]$ measuring the strength of support for an orthogonal interpretation of the tax change in period t. This shift moves the identifying argument away from hard classification and toward moment recovery: variation in p_t across observations is used to trace how contamination changes across the sample and to recover the orthogonal policy moment. The resulting estimator is therefore best understood as a moment estimator that adjusts for contamination rather than as a thresholding rule.

In that sense, the framework is closer in spirit to work on imperfect instruments and approximate moment conditions (Conley et al., 2012; Nevo & Rosen, 2012) and to recent analyses of inference under misspecified moment conditions (Armstrong & Kolesár, 2021; Bonhomme & Weidner, 2022; Hansen & Lee, 2021). The contribution here is to bring in a new source of auxiliary information: an LLM-based reading of policy documents that delivers an observation-level measure of narrative support. That measure disciplines how contamination varies across observations and thereby allows recovery of the orthogonal policy moment.

1.1 Traditional Narrative Identification as Hard Classification

Traditional narrative identification maps documentary evidence into a binary inclusion rule. A tax change is retained for identification when the historical record is judged to support an exogenous interpretation, and it is discarded otherwise. Let $\mathbb{1}\{\Delta\tau_t \text{ is classified as exogenous}\}$ denote that decision.

Although narrative studies typically report only the binary classification, it is useful to represent that decision as thresholding an underlying soft assessment $p_t \in [0,1]$:

$$\mathbb{1}\{\Delta\tau_t \text{ is classified as exogenous}\} \;=\; \mathbb{1}\{p_t \geq \bar{p}\},$$

for some evidentiary cutoff $\bar{p} \in (0,1)$. The identifying assumption of the traditional approach is that, once this cutoff is applied, the retained observations are treated as orthogonal to other macroeconomic disturbances:

$$\mathbb{E}[\varepsilon_{t+h}\Delta\tau_t \mid p_t \geq \bar{p}] = 0, \qquad \text{for } h \geq 0.$$

That is, narrative classification is used to construct a subset of tax changes that is taken to satisfy the exogeneity condition required for identification.

Under this interpretation, traditional narrative identification imposes two restrictions. First, it discards observations with intermediate narrative support even though these observations may still contain information about the relation between policy variation and macroeconomic outcomes. Second, once the threshold is crossed, residual classification uncertainty is ignored: retained observations are treated as providing orthogonal policy variation. The framework below relaxes both restrictions.

1.2 Soft Narrative Support and Contaminated Moments

Let $\mathcal{D}_t$ denote the contemporaneous documentary record associated with tax change t. Let $u_t \in \{0,1\}$ denote the latent class of the tax change, with $u_t = 1$ corresponding to the orthogonal class and $u_t = 0$ to the contaminated class. The documentary record is used to form a support index $p_t \in [0,1]$ for the claim that $u_t = 1$. A useful benchmark interpretation is to view p_t as approximating the posterior probability that the tax change belongs to the orthogonal latent class, conditional on the documentary record $\mathcal{D}_t$. The identifying argument itself requires calibration of the scalar support score:

$$\mathbb{E}(u_t \mid p_t = p) = \Pr(u_t = 1 \mid p_t = p) = p. \tag{1}$$

That is, among observations assigned support level p, the average fraction belonging to the orthogonal class is p.

Within-class moment stability and exclusion restriction. The affine benchmark also requires within-class moment stability: conditional on latent class membership, variation in narrative support should not systematically sort the average outcome-tax moment over the estimation sample. For each horizon $h \geq 0$ and all p in the support of p_t,[1]

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = j, p_t = p) = \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = j), \qquad j \in \{0, 1\}. \tag{2}$$

In particular, even if orthogonality holds within the orthogonal class, the cross moment may still vary with narrative support through systematic variation in shock magnitude, tax composition, implementation timing, anticipation, or related features within class. The benchmark also imposes the orthogonality restriction for the latent orthogonal class,

$$\mathbb{E}(\Delta\tau_t \varepsilon_{t+h} \mid u_t = 1) = 0, \qquad \text{for } h \geq 0.$$

Support-indexed moments. Under these restrictions, conditioning the outcome-tax moment on the support index gives:[2]

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p) = p\ \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) + (1-p)\ \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0). \tag{3}$$

Equation (3) makes the benchmark decontamination logic explicit. Under calibration and within-class moment stability, variation in p_t changes the posterior weight placed on the orthogonal and contaminated class moments. At the endpoint $p = 1$, the contaminated component receives zero weight, so the support-indexed moment coincides with the orthogonal-class moment. The object of interest is therefore $\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1)$. The corresponding normalized response is

$$\beta_h(1) = \frac{\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1)}{\mathbb{E}(\Delta\tau_t^2 \mid u_t = 1)}.$$

This normalizes the orthogonal-class numerator by the squared tax-change magnitude in the same latent class.

The expectations in this section should be interpreted as population counterparts of averages over the historical sample used in estimation. The framework does not require the macroeconomic environ-

[1] Appendix A shows that the same two-way representation can summarize a richer contaminated component, including multiple latent sources with opposite signs, provided the average contaminated moment is stable conditional on p_t.

[2] By iterated expectations over u_t, $\mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p) = \Pr(u_t = 1 \mid p_t = p)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1, p_t = p) + \Pr(u_t = 0 \mid p_t = p)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0, p_t = p)$. The calibration condition in equation (1) gives $\Pr(u_t = 1 \mid p_t = p) = p$ and $\Pr(u_t = 0 \mid p_t = p) = 1 - p$; the within-class moment stability restriction in equation (2) then allows the conditioning on p_t to be dropped inside the class-conditional moments.

ment to be stationary or identical across calendar time. More generally, class-conditional moments may vary across periods because of changes in tax composition, institutions, or macroeconomic regimes.

1.3 Identification from Variation in Narrative Support

The key source of identification is variation in p_t across dates. If p_t were constant, the data would reveal only one weighted average of the orthogonal and contaminated outcome-tax moments. With at least two distinct values of p_t, the weights vary and the orthogonal-class moment can be recovered.

To see this most clearly, suppose p_t takes two values, p_L and p_H, with $p_H \neq p_L$. Then (3) implies

$$\begin{cases} \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_L) = p_L\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) + (1 - p_L)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0), \\ \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_H) = p_H\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) + (1 - p_H)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0). \end{cases}$$

Solving this two-equation system gives the orthogonal-class moment:

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) = \frac{(1 - p_L)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_H) - (1 - p_H)\,\mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_L)}{p_H - p_L}.$$

Dividing this moment by $\mathbb{E}(\Delta\tau_t^2 \mid u_t = 1)$ gives the normalized response $\beta_h(1)$. The same system also identifies the contaminated-class moment, so both class-conditional moments are identified. The GMM formulation below preserves this affine case as the baseline, but writes the estimation problem in terms of the more general support-indexed moment.

1.4 Moment Conditions and a GMM Formulation

In practice, the moments are estimated after residualizing outcomes and tax changes on the controls included in the empirical specification.

More generally, let

$$m_h(p) \equiv \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p),$$

denote the support-indexed moment. For estimation, approximate this function by a finite-dimensional specification $m_h(p;\theta_h)$. Correct specification implies the conditional moment

$$\mathbb{E}[y_{t+h}\Delta\tau_t - m_h(p_t;\theta_h) \mid p_t] = 0. \tag{4}$$

Following the conditional-moment logic in Chamberlain (1987), equation (4) implies unconditional moments after multiplication by functions of p_t. In practice, choose a finite vector of functions $\phi(p_t)$ and impose $\mathbb{E}[(y_{t+h}\Delta\tau_t - m_h(p_t;\theta_h))\,\phi(p_t)] = 0$. The basis $\phi(\cdot)$ determines how much of the conditional restriction is used in estimation. A parsimonious basis gives a finite GMM approximation; richer bases move the estimator closer to the full support-indexed restriction.

Affine benchmark. Under the calibration and within-class moment-stability restrictions above, reparameterizing equation (3) gives the affine benchmark

$$m_h(p;\theta_h) = \underbrace{\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0)}_{a_h} + p\,\underbrace{[\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) - \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0)]}_{b_h},$$

with $\theta_h = (a_h, b_h)'$. Thus $a_h + b_h$ is the orthogonal-class moment and $\beta_h(1) = (a_h + b_h)/\mathbb{E}(\Delta\tau_t^2 \mid u_t = 1)$. For this baseline, the minimal just-identified estimator uses the two moments generated by $\phi(p_t) = (1, p_t)'$; equivalently, it is the OLS projection of $y_{t+h}\Delta\tau_t$ on a constant and p_t.

1.5 Measurement Error in Narrative Support

The preceding analysis treats the support index as observed, while in practice it is generated from text and may contain residual error from model choice, prompting, summaries, and ambiguity in the documents. Since p_t indexes the conditional moment $m_h(p)$, such errors affect both the location of an observation on the support scale and the finite GMM moments used to estimate the support-indexed function. The value of the GMM formulation is that, after projecting the conditional restriction on functions of p_t, measurement error can be studied as a perturbation of standard moment conditions.

The first benchmark is a residual classical-error approximation. After calibration and aggregation of the generated scores, corrected-moment methods in the spirit of Evdokimov and Zeleneev (2022) can be applied to the finite projected moments to understand the attenuation created by noisy support scores. This benchmark is useful because it gives a disciplined correction for small errors in the implemented GMM approximation.

The second layer is robustness to nonclassical error. Recent work on AI-generated and unstructured-data regressors emphasizes that generated variables can be model-dependent, biased, and correlated with features of the underlying text (Battaglia et al., 2024; Carlson & Dell, 2025; Ludwig et al., 2025). Repeated generated measures can document the stability of p_t and, under stronger independence assumptions, connect to multiple-proxy approaches such as Hu and Schennach (2008). More agnostically, one can allow the projected GMM moments to hold only approximately and ask how large the violation must be before the estimated orthogonal response changes materially, following bounded or approximate moment approaches (Armstrong & Kolesár, 2021; Conley et al., 2012).

2 Measuring Narrative Support for Exogeneity

Section 1 takes the support index as given. This section defines the measurement object behind that index and the restrictions imposed when constructing it from contemporaneous policy documents. The object is a policy-level scalar $p_i \in [0, 1]$ summarizing the extent to which the stated motivations for policy action i support assignment to the orthogonal narrative class. The procedure does not remove judgment from narrative identification. It makes that judgment explicit: the researcher must specify the relevant narrative classes, the mapping from stated motivations to those classes, the calibration of textual signals, and the aggregation rule from motivations to policy-level support. These choices become observable inputs to the empirical design rather than implicit features of a binary classification.

I write p_i for the policy-level support score used in the paper. When comparing admissible measurement specifications, $\hat{p}_i^{(r)}$ denotes the score produced by specification r. Although the empirical procedure starts from documents, I first define the probability object constructed from stated motivations, since this is the object that enters the identification argument. I then describe how those motivations are recovered from the documentary record. The document-to-support mapping is

$$(q_i, \mathcal{D}_i) \to \mathcal{R}_{K,L}(q_i; \mathcal{D}_i) \to \mathcal{M}_i \to \{p_{i\ell}\}_{\ell \in \mathcal{M}_i} \to p_i.$$

What is required is a semantic mapping from stated motivations into the latent classes relevant for identification, such as orthogonal versus contaminated, or exogenous versus endogenous. In the applications below, I adopt the mapping provided by the narrative tax literature, following Romer and Romer (2010)'s distinction between tax changes motivated by current economic conditions and those motivated by more orthogonal objectives, and the more detailed taxonomy proposed by Cloyne (2013). The contribution of the framework is therefore not to redefine the narrative notion of exogeneity, but to quantify the extent to which the documentary record supports assigning a given policy action to that narrative class. The details of this tax-policy implementation are described in Section 3.

2.1 The Policy-Level Support Object

For a policy action i, the measurement object is the policy-level probability p_i that the contemporaneous documentary record supports interpreting the action as belonging to the orthogonal latent class. The primitive textual evidence is the set of stated motivations recovered from that record,

$$\mathcal{M}_i = \{M_{i1}, \ldots, M_{iL_i}\}.$$

A motivation is a short statement of the reason given for the policy action. The motivation-level object is the support probability $p_{i\ell}$, and the policy-level object is a scalar aggregation $p_i \in [0, 1]$.

This construction preserves mixed narrative evidence. A policy action may contain several stated motivations, some pointing toward the orthogonal class and others toward the contaminated class. The object p_i records the strength of support for the orthogonal interpretation rather than forcing the action into a binary narrative label.

2.2 From Stated Motivations to Support Probabilities

Given stated motivations, the empirical task is to assign calibrated probability mass over the researcher-supplied narrative taxonomy. Let $\mathcal{G}$ denote the set of narrative categories and let $\mathcal{G}_{\text{exo}} \subset \mathcal{G}$ denote the categories interpreted as supporting the orthogonal class. The categories are an intermediate representation: the object needed for identification is the probability mass assigned to the orthogonal block.

The 1977 French VAT reform illustrates the mapping. This is a useful example because it is a major tax change: the reform reduced the standard VAT rate from 20 percent to 17.6 percent, merged it with the intermediate rate, and carried a reported fiscal impact of 8.5 billion francs, about 0.5 percent of GDP. It is also useful because the contemporaneous record contains both short-run stabilization motives and structural-reform motives; the same mapping is applied to the U.S. and UK applications below. The retrieved documentary evidence is condensed into three stated motivations:

- ***Mitigate the rise in prices after the end of the price freeze****: the measure aims to limit the inflationary rebound likely to occur at the end of the price-freeze period by reducing the tax burden on consumption.*

- ***Complete a structural reform****: the measure is the final step in a reform undertaken since 1969 to merge the standard VAT rate with the intermediate rate and reduce the number of VAT rates.*

- ***Fight inflation***: *the reduction of the rate is part of the broader government programme to combat inflation by easing indirect taxation.*

These motivations are then mapped into category probabilities and aggregated into a policy-level support probability.

Category probabilities. For each motivation, the measurement task is to construct calibrated probability mass over $\mathcal{G}$. In the baseline implementation, these probabilities are obtained from language-model representations of the motivation, following recent work that uses model representations and classifier probabilities as statistical measurements of text (Buckmann & Hill, n.d.; Chen et al., 2026; Hartley, 2025). Let $h(M_{i\ell})$ denote the hidden-state representation of motivation $M_{i\ell}$, and let κ denote a calibrated statistical map from hidden states to category probabilities:

$$\hat{P}_{i\ell,c} := \kappa_c(h(M_{i\ell}))\,, \qquad c \in \mathcal{G}. \tag{5}$$

The vector $(\hat{P}_{i\ell,c})_{c\in\mathcal{G}}$ is normalized to sum to one. In the empirical implementation, these probabilities are averaged across calibrated specifications that differ in model choice and calibration design. Section 3.3 reports the concrete calibration datasets, models, and aggregation choices used in the applications, including the multinomial logistic calibration used in the baseline specification. Appendix F records an alternative implementation based on incremental log probabilities.

Aggregation to policy-level support. The motivation-level probability of supporting the orthogonal class is the calibrated probability mass assigned to exogenous categories:

$$\hat{p}_{i\ell} = \sum_{c\in\mathcal{G}_{\text{exo}}} \hat{P}_{i\ell,c}. \tag{6}$$

For a policy action with extracted motivations $\mathcal{M}_i = \{M_{i1}, \ldots, M_{iL_i}\}$, the baseline policy-level support probability is the equal-weight average of motivation-level support:[3]

$$p_i = \frac{1}{L_i}\sum_{\ell=1}^{L_i} \hat{p}_{i\ell}. \tag{7}$$

Table 1 reports the calibrated category probabilities for the VAT example. The first and third motivations load mainly on Demand Management, so they contribute mostly to the endogenous or contaminated block. The second motivation loads more on categories in the exogenous block, including Ideological, Long-Run, and Deficit Consolidation. The resulting policy-level support probability is therefore interior: the displayed calculation assigns 0.35 of the probability mass to the exogenous block and 0.65 to the endogenous block.

[3] More general aggregation rules could replace the equal-weight average with $p_i = \sum_{\ell=1}^{L_i} \omega_{i\ell}\hat{p}_{i\ell}$, where $\sum_{\ell=1}^{L_i} \omega_{i\ell} = 1$. Such weights should be used only when the researcher has an auditable rule for ranking the relative importance of stated motivations.

Table 1: Robust calibrated category probabilities for the 1977 VAT reform example

	Category	M1	M2	M3	Overall policy prob.
Endo	Demand Management (DM)	0.75	0.02	0.69	0.65
	Spending-Driven (SD)	0.02	0.09	0.04	
	Supply Stimulus (SS)	0.01	0.01	0.01	
	Deficit Reduction (DR)	0.17	0.05	0.08	
Exo	Deficit Consolidation (DC)	0.02	0.22	0.05	0.35
	External (EXT)	0.01	0.05	0.05	
	Ideological (IL)	0.01	0.35	0.04	
	Long-Run (LR)	0.01	0.22	0.05	

Note: Each motivation column sums to 1 across the eight categories. The shaded cell marks the highest category within each motivation. The final column reports the policy-level probability mass assigned to the endogenous and exogenous blocks after averaging across motivations; the two block probabilities sum to 1.

2.3 Recovering Stated Motivations from Documents

The preceding subsection takes stated motivations as given. I now describe how they are recovered from the documentary record. Let q_i be a short policy description and let $\mathcal{D}_i$ denote the contemporaneous document collection associated with policy action i. Partition the document collection into segments of length L,

$$\mathcal{D}_i = \{d_{i,1}, \ldots, d_{i,N_i}\}.$$

A text representation $\psi(\cdot)$ maps the policy description and each document segment into a common semantic space. Relevance is measured by a distance

$$s_{i,k} = \operatorname{dist}\{\psi(q_i), \psi(d_{i,k})\}, \qquad d_{i,k} \in \mathcal{D}_i.$$

The retrieval operator selects the K closest passages,

$$\mathcal{R}_{K,L}(q_i; \mathcal{D}_i) = \underset{d_{i,k}\in\mathcal{D}_i}{\overset{K}{\arg\min}}\, s_{i,k} = \{d_i^{(1)}, \ldots, d_i^{(K)}\}. \tag{8}$$

This semantic-distance retrieval step is part of the measurement mapping. The concrete embedding model, segment length, and retrieval depth are implementation choices described in Section 3.

The retrieved passages are then condensed into stated motivations. Formally, a summarization operator maps the retrieved evidence into a structured set of motivations,

$$\mathcal{S}\{\mathcal{R}_{K,L}(q_i; \mathcal{D}_i)\} = \mathcal{M}_i.$$

The purpose of this step is constrained evidence compression: the summarization operator is not allowed to infer motives absent from the retrieved passages, but only to express stated rationales in a compact and comparable form. In the VAT example, this retrieval and summarization step produces the three motivations used in the probability calculation above.

2.4 Measurement Error, Repeated Measures, and Traceability

The support score is generated from text and should be treated as a measured input rather than as an error-free observation. Conditional on the researcher's measurement choices, however, the mapping from documents to support is deterministic and systematic. Let

$$\eta = (\eta_R, \eta_S, \eta_C, \eta_K, \eta_A)$$

collect the choices governing retrieval, summarization, semantic classification, calibration, and aggregation. A given admissible specification corresponds to a value $\eta^{(r)}$ and returns

$$\hat{p}_i^{(r)} = f(q_i, \mathcal{D}_i; \eta^{(r)}).$$

When repeated specifications are used, the reported support score is the average

$$p_i = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \hat{p}_i^{(r)}. \tag{9}$$

Differences across $\hat{p}_i^{(r)}$ capture sensitivity to measurement choices. Relevant evidence may be missed at retrieval, summarized motivations may omit or overstate parts of the documentary record, category probabilities may be miscalibrated, and aggregation may place the wrong weight on competing motivations. Agreement across specifications is informative, but it is not sufficient for validity: a stable score may still reflect generic language patterns rather than policy-specific evidence. For this reason, Section 3.6 reports diagnostics that assess whether extracted motivations are grounded in the retrieved policy-specific passages.

Traceability is an audit requirement. Each final support score should be linked to the policy description, source passages, extracted motivations, category probabilities, calibration choices, and aggregation rule used to transform textual evidence into probabilities. This subsection defines the measurement logic. Section 3 specifies the particular choices used in the applications. The point is not that a given language-model implementation is definitive, but that the object entering the econometric analysis is auditable, replicable conditional on stated choices, and open to sensitivity analysis.

The object defined here is action-level support. When the econometric analysis is conducted at quarterly frequency, these action-level scores are aggregated into p_t using the amount-weighting rule described in Section 3.4.

3 Common Empirical Implementation

This section records the common empirical implementation used to construct narrative-support probabilities in the tax applications. Section 2 defined the object p_i and the generic document-to-support mapping. This section specifies the choices used in the current tax applications: the tax-policy taxonomy, document-processing settings, calibration design, action-level aggregation, econometric conventions, and diagnostics. All applications use this common implementation unless otherwise stated.

Group	Sub category	Explanation and examples
Endogenous (N)	1. Demand management (DM)	• Targeting the aggregate level of demand e.g. to boost investment, consumption, growth, or curb inflation. • Specific help to households and individuals by stimulating disposable income. • Dealing with a balance of payments crisis via demand.
	2. Supply stimulus (SS)	• Certain help for businesses during a downturn (e.g. NIC cut). • Short term sector support (e.g. targeted tax cuts for a sector).
	3. Deficit reduction/BoP crisis (DR)	• Direct measures to deal with a budget or external deficit contemporaneously caused.
	4. Spending-driven (SD)	• Taxes which fund specific spending commitments.
Exogenous (X)	1. LR performance (LR)	• Measures to improve competitiveness, productivity, efficiency and long-run growth (but not taken to offset a shock). • Simplification and deregulation measures. • Long-term support for business or sectors of the economy.
	2. Ideological (IL)	• Long-term social or political goals, independent of their effect on performance and not to offset current shocks. • Some anti-avoidance measures (where no other motive is given).
	3. External (ET)	• Court rulings and enforcement of directives.
	4. Deficit consolidation (DC)	• Measures to lower inherited deficit for reasons of economic philosophy or to offset current actions in the future. • Does not include actions forced on the government, or decisions contemporaneous motivated by a current shock.

Figure 1: Classification of Tax Measures Following Cloyne (2013)

3.1 Semantic Taxonomy for Tax-Policy Motivations

For tax policy, the generic semantic mapping is anchored in the narrative tax literature. Romer and Romer (2010) distinguish tax changes motivated by current economic conditions from those motivated by more orthogonal objectives. Cloyne (2013) refine this distinction into a set of tax-policy motivation categories. I use this taxonomy because it translates the abstract latent classes in Section 1 into auditable economic rationales for tax changes.

Table 1 recalls the Cloyne categories. Endogenous measures are those linked to contemporaneous economic or fiscal conditions, such as demand management, supply stimulus, deficit reduction, or spending-driven motives. Exogenous measures are justified by objectives less directly tied to the current state of the economy, such as long-run reform, ideological objectives, external constraints, or deficit consolidation. This taxonomy is an implementation choice for the tax applications, not part of the abstract measurement object.

Each extracted motivation is assessed against the category definitions using category-specific prompts. The prompts are designed to return an auditable category signal and a short rationale tied to the stated motivation. The prompt templates are reported in Appendix D.2.

3.2 Document Processing and Model Choices

Retrieval settings. The implementation uses OpenAI's *text-embedding-3-large* model to locate relevant passages. For each policy action i, the policy description is used as the query against the associated document collection. Documents are split into fixed-length chunks and the baseline retrieval keeps the ten most relevant chunks. Retrieval quality is checked on the French VAT corpus using a manually constructed *gold article* benchmark: each VAT measure is linked to the Senate-report article that discusses the corresponding budget-law provision. Across chunk lengths $L \in \{512, 1024, 2048, 4096\}$, retrieval is highly stable. For $L = 2048$, at least one of the top-ten chunks overlaps the gold article in 97 percent of cases, and on average 5.5 of the ten retrieved chunks overlap the gold article.

Summarization and classification settings. The retrieved passages are passed to the language models with a fixed prompt asking for the government's stated motivations for the tax change. The prompt instructs the model to rely only on the retrieved excerpts, avoid adding interpretation, and return at most three motivations in a uniform bullet-point format. The dynamic fields in the prompt are the fiscal year, the tax type, and the policy description taken from the policy dataset.

The summarization and classification stages are performed with three open-weight instruction-tuned models: `Gemma-4B`, `Mistral-8B-Instruct`, and `Qwen-3.5-9B`. Stochastic elements of text generation are disabled. All retrieved passages, prompts, outputs, model identifiers, hidden-state files, and decoder settings are stored so that each final support score can be traced back to the source text, generated motivations, calibration set, and model calls.

3.3 Calibration and Robust Probability Construction

Section 2.2 describes the hidden-state calibration logic at a generic level. The current robust application calibrates the probability scale with a synthetic motivation corpus rather than real policy labels. The corpus uses the same eight-category taxonomy as the tax application, but the generation prompt is separate from the downstream classification prompt. The current calibration design contains twelve independently generated sets: four prompt families times three random seeds. The families stress different parts of the classifier: clear no-template examples, U.S./Romer–Romer-style institutional language, hard negatives with misleading cue words, and mixed-motive examples with one primary category. Each set has 60 examples in each of the eight categories, so every set contains 480 balanced calibration rows. Real motivations are used only for coverage and transfer diagnostics, not for training the calibration decoder.

For each synthetic set k and each open-weight model m, the implementation computes motivation-only hidden states and trains an eight-class multinomial logistic decoder. The regularization parameter is selected by cross-validation inside the synthetic set. The trained decoder returns a probability over the eight narrative categories for each real motivation. Let $\mathcal{E}$ denote the set of exogenous categories. For motivation j in policy action i, the model–set-specific exogeneity probability is

$$p_{ij}^{(k,m)} = \sum_{c \in \mathcal{E}} \widehat{\Pr}_{k,m}(c \mid h_{ij}^{(m)}),$$

where $h_{ij}^{(m)}$ is the hidden-state representation from model m. When an action has multiple extracted motivations, the action-level probability first averages over motivations. It then averages over the three open-weight models and over the twelve synthetic calibration sets,

$$p_i^{\text{rob}} = \frac{1}{12}\sum_{k=1}^{12} p_i^{(k)}, \qquad p_i^{(k)} = \frac{1}{3}\sum_{m \in \mathcal{M}} p_i^{(k,m)}, \quad \mathcal{M} = \{\texttt{Gemma}, \texttt{Mistral}, \texttt{Qwen}\}.$$

The cross-set standard deviation, together with the 10th and 90th percentiles across set-trained decoders, is stored as calibration uncertainty. The empirical column used for the current robust application is `p_robust_synthetic_hidden_cat8_v1`.

3.4 Action-Level to Quarterly Aggregation

The econometric input is an action-level dataset with three required objects: an implementation date, a scaled signed tax amount τ_i, and the support score $p_i \in [0,1]$. The date determines the quarter in which the action enters the macroeconomic calendar. The tax amount is expressed in the same normalization as the country-specific tax-shock series, for example as a percent of GDP or trend GDP.

Let $\mathcal{A}_t$ denote the set of policy actions assigned to quarter t. The quarterly tax change is the signed sum of action-level tax changes,

$$\tau_t = \sum_{i \in \mathcal{A}_t} \tau_i.$$

The quarterly support score is an absolute-tax-weighted average of action-level support scores,

$$p_t = \frac{\sum_{i \in \mathcal{A}_t} |\tau_i| p_i}{\sum_{i \in \mathcal{A}_t} |\tau_i|}, \qquad \text{if } \sum_{i \in \mathcal{A}_t} |\tau_i| > 0.$$

This weighting gives more influence to the narrative support attached to larger tax changes while allowing tax increases and tax cuts in the same quarter to offset in the net tax shock. If a quarter contains no action, p_t is set to zero by convention. Quarters with positive gross action mass but exactly offsetting signed changes are retained in construction diagnostics.

3.5 Econometric Implementation

The country-specific sections define the macro sample, the outcome variables, the horizon grid, and the exact control vector. For each horizon h, the calendar sample $\mathcal{T}_h$ contains quarters for which the lead outcome y_{t+h} and controls $W_{t,h}$ are observed. The controls are quarterly predetermined variables, typically lags of the outcome, other macro variables, and the tax-shock series.

Residualization. For each horizon h, the implementation residualizes the outcome lead and the quarterly tax shock before forming the GMM numerator:

$$\tilde{y}_{t,h} = y_{t+h} - \widehat{\pi}'_{y,h} W_{t,h}, \qquad \tilde{\tau}_{t,h} = \tau_t - \widehat{\pi}'_{\tau,h} W_{t,h}.$$

The implementation keeps the original quarterly ordering when forming inference scores, so that HAC covariances preserve the calendar spacing of the shocks. The numerator object is

$$x_{t,h} = \tilde{y}_{t,h} \tilde{\tau}_{t,h}.$$

Affine GMM projection. For each horizon h, the baseline affine implementation estimates the numerator moment with the horizon-specific projection

$$x_{t,h} = \alpha_h + \lambda_h p_t + u_{t,h}.$$

The affine numerator evaluated at support level p is $M_h(p) = \alpha_h + \lambda_h p$. The reported high-support endpoint uses $M_{A,h} = M_h(1) = \alpha_h + \lambda_h$, with $M_{0,h} = M_h(0) = \alpha_h$ as the corresponding low-support diagnostic.

Denominator normalization. The paper-facing endpoint response uses the denominator evaluated at the same high-support endpoint as the numerator. The implementation estimates the affine denominator projection

$$\tilde{\tau}_{t,h}^2 = \alpha_{z,h} + \lambda_{z,h} p_t + v_{t,h},$$

and defines

$$Q_{A,h} = \alpha_{z,h} + \lambda_{z,h}.$$

The headline endpoint response is

$$\widehat{\beta}_{A,h} = \frac{\widehat{M}_{A,h}}{\widehat{Q}_{A,h}}.$$

Alternative specifications and inference. The implementation stores diagnostic variants that replace the affine basis with hard thresholds, bins, rank-normalized affine specifications, local linear specifications, and splines. These alternatives check whether results are driven by extrapolating a linear support-index relationship to $p = 1$. Standard errors are based on time-series HAC covariance matrices applied to the full-calendar influence scores. The bandwidth is at least four quarters and increases with the horizon, using $\max(4, h)$ in the baseline implementation. Ratio standard errors use the delta method and include sampling variation in the endpoint denominator.

3.6 Reporting and Robustness Diagnostics

The implementation reports diagnostics for the text-generated probabilities, the action-to-quarter aggregation, the GMM projection, and the document grounding of generated motivations. These diagnostics are stored with the IRF output so that changes in estimated responses can be traced back to changes in the support score, the quarterly tax-shock construction, the numerator moment, or the denominator normalization.

Model agreement. Because the final probability averages over open-weight models, the implementation reports agreement across the three model-specific probabilities. In the current output, the average pairwise Pearson correlations are about 0.82–0.83 in the U.S. samples and about 0.72 in the France samples; the corresponding average Spearman correlations are similar. Median within-row model standard deviations are roughly 0.09–0.11. The same output reports cross-set stability: median set-level standard deviations are about 0.09–0.12 across the U.S. and France samples.

Construction diagnostics. For each probability specification and sample, the implementation reports the number of usable quarters, action counts, the distribution of action-level and quarterly p, signed and absolute tax-shock mass, variation in p_t, denominator diagnostics, rank checks, and flags for near-zero denominators or insufficient variation. Retrieval quality is checked in applications where a manual benchmark is available; for the French VAT corpus, Appendix D.1 reports a gold-article retrieval diagnostic.

Table 2: Model agreement after averaging over synthetic calibration sets

Sample	N	Mean Pearson	Mean Spearman	Median model s.d.
U.S. RR replication	50	0.82	0.80	0.09
U.S. extension	1188	0.83	0.84	0.11
France VAT	140	0.72	0.72	0.10
France non-VAT	1020	0.72	0.74	0.11

4 External Validation and U.S. Application

This section validates the proposed framework and then applies it to an end-to-end U.S. budget-document exercise. The empirical design separates three questions. First, using edited Romer–Romer narratives, does the probability-construction stage assign higher support to episodes classified as exogenous in the canonical U.S. narrative series? Second, using Cloyne's UK tax data, does the full document-grounded pipeline align with an independent expert classification when retrieval, summarization, and motivation extraction are all performed from policy documents? Third, using routine U.S. budget measures outside the original Romer–Romer sample, do document-based probabilities generate tax shocks with macroeconomic content comparable to the benchmark series?

This sequencing deliberately separates probability validation from macroeconomic application. The Romer–Romer and Cloyne exercises evaluate whether the probability object is meaningful relative to expert narrative classifications. The final U.S. budget exercise then asks whether that object can expand identifying support in a setting where no complete expert-coded classification exists for the newly added measures.

4.1 Validation using the Romer–Romer narratives

This subsection evaluates whether the common implementation in Section 3 recovers the economic content of Romer and Romer (2010) when it is applied to the edited narratives used in the original study. The goal is not to reinterpret the historical record or to force an automated binary relabeling of the RR episodes. Instead, the exercise asks whether the procedure introduced in the paper assigns systematically higher exogeneity support to RR-exogenous episodes and whether the resulting probability-based shock series reproduces the benchmark macroeconomic response.

By conditioning on the curated Romer–Romer narratives, this benchmark abstracts from document retrieval while still preserving the summarization, motivation decomposition, and probability aggregation stages of the framework. It therefore provides a test of whether the probability object constructed by the methodology carries a similar identifying content as the original RR narratives.

4.1.1 Setup

The Romer–Romer replication material documents 50 major federal tax changes in the United States between 1945 and 2007. For each episode, the authors provide a detailed narrative discussion of the motivations underlying the reform, drawing on contemporaneous sources such as presidential speeches, budget documents, and Congressional records. Each narrative concludes with a classification of the tax change as either exogenous or endogenous.

To construct the benchmark validation task, I extract the narrative text describing each policy episode and remove the portion in which the authors explicitly state and justify their classification. In particular, I delete the sentence reporting the final classification along with the two immediately preceding sentences, which often contain direct cues revealing the intended label (see Appendix E.1.1 for examples). The remaining narrative text is used as input to the language model.

4.1.2 From edited narratives to policy-level probabilities

The edited RR narrative does not enter the benchmark as a single paragraph to be mapped directly into a binary label. Instead, each edited narrative is first summarized into a set of stated motivations. Because a single RR episode can contain several rationales, the same policy may generate multiple motivations, each of which is scored separately. These motivation-level outputs are then mapped into exogeneity probabilities and aggregated into a policy-level structural probability.

The construction follows the cross-implementation aggregation described in Subsection 3.6; see in particular equation (9). Throughout this benchmark, the object of interest is therefore the policy-level narrative support measure p_i, rather than any single implementation-specific estimate $\hat{p}_i^{(r)}$.

4.1.3 Probability validation against RR classifications

Table 3 reports the original Romer–Romer binary label and asks whether RR-exogenous episodes receive systematically higher policy-level probabilities. The mean probability rises from 0.40 for RR-endogenous episodes to 0.62 for RR-exogenous episodes, and the share above tighter thresholds also rises.

Table 3: Policy-Level Structural Probabilities by Original RR Classification

RR class	**N**	**Mean** p	**Share** $p \geq 0.5$	**Share** $p \geq 0.75$
RR endogenous	24	0.40	20.8%	0.0%
RR exogenous	26	0.62	69.2%	34.6%
Difference (exo - endo)	–	0.23	48.4 pp	34.6 pp

Note: Entries are computed from the 50 edited RR narratives. p is the robust policy-level structural probability averaged across independently generated synthetic motivation calibration sets.

The separation is informative but far from complete, and that is precisely why a continuous object is useful. The relevant question is not whether the procedure reproduces a single RR label exactly, but whether episodes judged exogenous by Romer and Romer receive systematically more documentary support for exogeneity. Under a probability-based implementation, ambiguous cases are down-weighted rather than being forced into an all-or-nothing inclusion rule. Appendix E.2 provides probabilistic label-prediction and training-leakage checks.

4.1.4 Macroeconomic implications

I next assess whether these policy-level probabilities preserve the macroeconomic content of the Romer–Romer shock series. For GDP, I now match the published RR benchmark exactly: quarterly real GDP growth is regressed on the contemporaneous tax shock and twelve lags of that shock, together with eleven lags of GDP growth, over the 1950Q1–2007Q4 sample. The reported output response is the implied cumulative effect on GDP, and all curves are shown with 68% confidence bands. This subsection focuses on GDP because that is the benchmark specification reported in Romer and Romer (2010).

Figure 2 compares the original RR series to two estimators built from the policy-level probability p. Panel (a) reports the hard-threshold estimator with $\bar{p} = 0.5$, which converts the continuous support measure back into a binary-style filter. Panel (b) reports the two-type estimator evaluated at $p = 1$, corresponding to the most exogenous endpoint implied by the affine mixture specification.

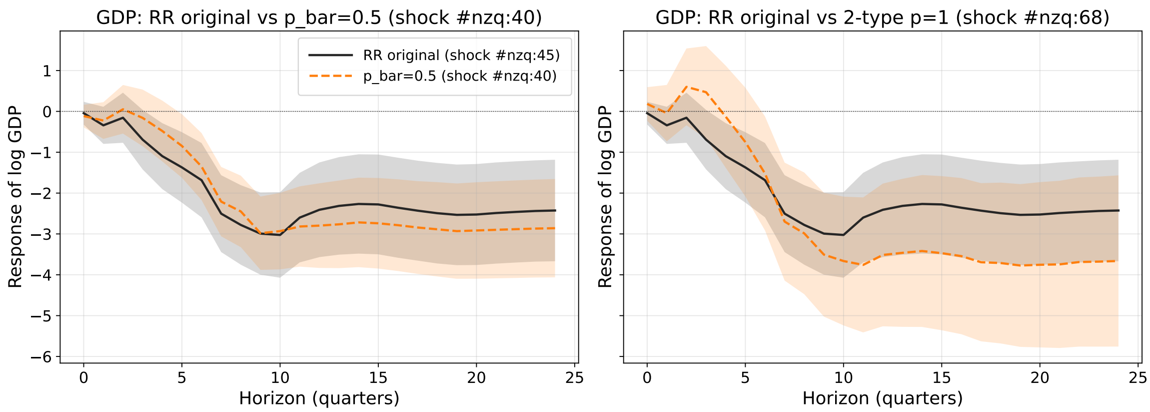

Figure 2: Romer–Romer GDP Responses: Original Series versus Probability-Based Estimators

Note: Each panel reports the exact Romer–Romer GDP specification. The black line is the original RR exogenous shock series. The colored line is the corresponding estimator constructed from `p`. Shaded areas denote 68% confidence bands.

The two panels are informative in different ways. The original RR series reaches a peak contraction of roughly 3 log points around quarter 10, in line with the original Romer–Romer result. The hard-threshold estimator asks whether the probability can be converted back into a binary-style exogeneity rule. The two-type $p = 1$ endpoint instead uses the probability ranking to estimate the high-support endpoint of an affine mixture. Both comparisons remain much closer to the canonical RR response than the series obtained by treating all tax changes as equally exogenous.

Figure 3 unpacks the two-type estimator more directly. The left panel reports the two endpoint responses, $p = 0$ and $p = 1$. The right panel then shows the implied linear mapping at horizon $h = 10$, which is where the exact-RR benchmark reaches its peak contraction in the current implementation. Read this way, the two-type estimator is tracing how the estimated response moves between the low-support and high-support endpoints as the policy-level exogeneity probability increases. In the RR benchmark, the gap between the two endpoints is already substantial by $h = 10$, with the high-p endpoint much closer to the canonical RR contraction than the low-p endpoint.

Taken together, these results show that policy-level exogeneity probabilities extracted from edited RR narratives recover the central macroeconomic content of the original benchmark. If moving toward higher-p tax changes systematically pulls the estimated response toward the RR benchmark and away from the attenuated all-tax-changes series, then the probability pipeline is capturing the same identifying content as the RR narratives themselves. The next subsection asks whether this probability interpretation also survives a full document-pipeline validation outside the United States.

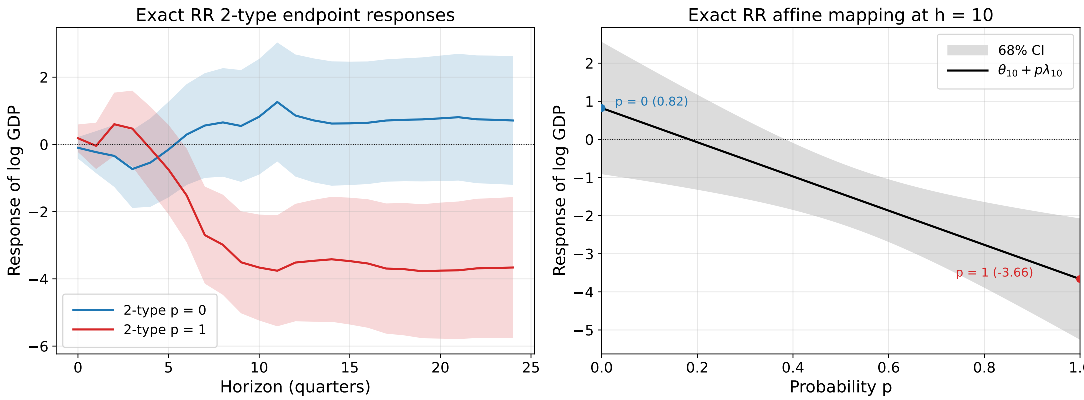

Figure 3: Two-Type RR Benchmark: Endpoint IRFs and the Mapping at $h = 10$

Note: The left panel reports the exact-RR two-type endpoint responses for $p = 0$ and $p = 1$, each with 68% confidence bands. The right panel plots the affine response mapping $\theta_h + p\lambda_h$ at horizon $h = 10$, together with its 68% confidence band.

4.2 External Label Validation: Cloyne (2013) UK Tax Changes

The Romer–Romer exercise above isolates the probability-construction stage by starting from curated narrative text. I next ask whether the same probability object remains informative in a more demanding setting, where the relevant policy documents must first be reconstructed. Cloyne (2013) provides a useful benchmark for this purpose. The dataset records individual UK tax changes, their revenue effects and implementation dates, and an expert narrative classification of each change as endogenous or exogenous.

The validation starts from Cloyne's row-level tax workbook. The reconstruction preserves the 2,462 tax-change rows in the original file and maps them into 85 narrative events. For each event, I build an event-level source base from contemporaneous UK fiscal and parliamentary materials, apply the retrieval and summarization procedure described above, and extract up to three stated motivations for each policy row. These motivations are then mapped into structural exogeneity probabilities. Because Cloyne's classification is also defined at the policy-row level, it provides a direct external label against which to compare the resulting probabilities.

This exercise is a measurement validation, not a UK macroeconomic application. UK output, inflation, and revenue responses are not used to construct the probabilities. The question is instead whether rows classified as exogenous by Cloyne receive more documentary support for exogeneity than rows classified as endogenous. Table 4 presents the comparison in the same spirit as the Romer–Romer validation table.

The separation is positive but more modest than in the Romer–Romer validation. The updated probabilities are generally high for both Cloyne classes, so the 0.5 cutoff is not very discriminating. The ranking is nevertheless monotone: Cloyne-exogenous rows have an average probability of 0.76, compared with 0.69 for Cloyne-endogenous rows, and the share above the 0.75 threshold is 19.8 percentage points higher. Appendix E.2.2 confirms the same monotone relationship in a one-variable probabilistic label-prediction exercise.

The alignment is nevertheless imperfect. Many Cloyne-endogenous rows receive fairly high probabil-

Table 4: Policy-Level Structural Probabilities by Cloyne Classification

Cloyne class	**N**	**Mean** p	**Share** $p \geq 0.5$	**Share** $p \geq 0.75$
Cloyne endogenous	795	0.69	88.2%	40.5%
Cloyne exogenous	1,306	0.76	95.8%	60.3%
Difference (exo - endo)	–	0.08	7.6 pp	19.8 pp

Note: Entries are computed from 2,101 policy rows with usable Cloyne labels and complete probability inputs. "Cloyne exogenous" denotes rows with Cloyne's major classification X. p is the policy-level structural probability used in the UK validation exercise. No UK macroeconomic outcome enters the probability construction.

ities, often because the reconstructed source text contains mixed package-level motivations or because Cloyne's policy-row classification aggregates several motives within a fiscal package. This pattern is useful rather than problematic for the present purpose. It shows why the object of interest is a continuous measure of support, not a forced reproduction of an expert binary label. It also clarifies the role of the following U.S. application: after validating the probability object in a curated-narrative setting and in a full-document setting with external labels, I use it to expand identifying support where no complete expert-coded benchmark exists.

4.3 End-to-end extension using U.S. policy documents

The preceding subsections validate the probability object against expert narrative classifications: first using curated Romer–Romer narratives, and then using Cloyne's full UK policy-document setting. I now turn to the end-to-end U.S. application. The objective is to ask whether newly constructed budget-based shocks become more RR-like once documentary support for exogeneity is used to filter or weight routine federal tax changes that fall largely outside the original Romer–Romer sample.

4.3.1 U.S. tax data and coverage

Existing narrative coverage. The narrative tax dataset of Romer and Romer (2010) identifies major federal tax actions in the United States over the postwar period, focusing on large and salient legislative episodes. The series includes comprehensive tax reforms and major standalone acts, such as income tax reforms, Social Security legislation, and federal highway funding acts, that generate sizable and clearly identifiable changes in aggregate tax liabilities. By construction, it abstracts from routine fiscal adjustments and from smaller tax provisions embedded in annual budgetary processes. As a result, the RR series provides a measure of large, infrequent tax shocks, but it offers only partial coverage of legislated tax changes and at an aggregated tax-instrument level.

Extension. To recover the tax changes omitted, I construct a comprehensive dataset of legislated federal tax measures by systematically processing each year's *Budget of the United States Government* from 1950 to 2024. For each fiscal year, I extract all reported changes in federal taxation and record them at the highest level of detail available, including a disaggregation by tax instrument whenever possible. The same extraction procedure is applied to the major tax acts identified by Romer and Romer (2010), yielding a finer tax-level decomposition than that available in the extension of Mertens and Ravn (2013).

For the end-to-end validation below, I focus on the budget-based portion of this broader dataset. That choice is deliberate: the annual budget volumes provide a large set of routine tax measures that mostly fall outside the canonical RR benchmark and therefore furnish an out-of-sample environment in which to test whether the probability-based procedure recovers economically meaningful identifying content.

Table 5 reports the composition of the resulting budget dataset. It aggregates all tax changes explicitly proposed in the yearly *Budget of the United States Government* over the sample period and classifies them into six major federal tax categories. The table therefore describes the structure of tax policy adjustments as they appear in executive budget proposals, not the universe of all federal tax legislation. In the final dataset, each tax change is associated with the quarter in which it becomes effective. In particular, when no specific quarter is mentioned, the tax change is assigned by default to the first quarter of the year.

Table 5: Distribution of New Tax Measures by Category (United States)

Tax Category	**# New Measures**
Individual Income Tax	371
Corporate Income Tax	361
Consumption Tax	238
Payroll / Social Insurance Tax	114
Other Receipts	89
Wealth Transfer Tax	17
Total	**1190**

4.3.2 Document corpus

A final step in constructing the U.S. tax dataset is to assemble a corpus of contemporaneous documents that records policymakers' stated motivations for legislated tax changes. Sources are selected to reflect the policy rationale articulated at the time of enactment and to remain consistent and comparable over long horizons.[4] The broader U.S. corpus relies on two recurring source families: the *Budget of the United States Government* and the *Annual Report of the Secretary of the Treasury*.

For the end-to-end benchmark in this subsection, the relevant document source is the annual Budget itself. These budget volumes are the full-document analogue of the edited RR narratives used above: instead of starting from hand-curated narrative excerpts, the procedure retrieves and summarizes contemporaneous budget text. The application follows the common implementation in Section 3; additional technical details are provided in Appendix B.4.

4.3.3 Distribution of estimated exogeneity support

Figure 4 reports the distribution of the policy-level probability p across the budget measures that enter the quarterly shock construction. Unlike the earlier binary benchmark, the distribution is overwhelm-

[4] Romer and Romer (2010) draw on a broad set of contemporaneous sources, including legislative debates and discussions in congressional chambers. Some of these materials are not systematically used here, primarily because their access and coverage are uneven over time. The analysis therefore focuses on recurring executive and Treasury documents that are consistently available throughout the sample period.

ingly interior. In the current matched sample, the mean is about 0.55, the median is about 0.59, 61% of measures have $p \geq 0.5$, and 33% have $p \geq 0.7$, while neither exact zero nor exact one occurs once multiple motivations are aggregated at the policy level.

This is precisely the environment in which a probability object is useful. Routine budget measures often combine several stated motivations, so the aggregation step naturally produces interior support rather than literal corner solutions. The right panel of Figure 4 makes that filtering margin transparent: moving from all measures to high-support measures is not a matter of a few isolated corner cases, but of filtering a wide interior distribution.

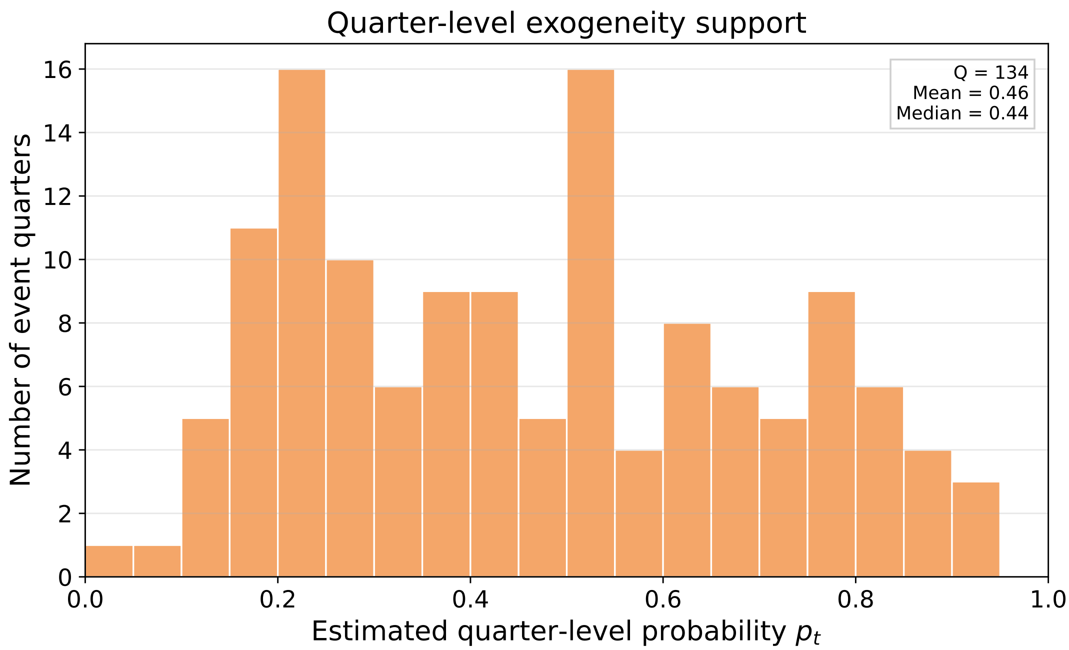

Figure 4: Distribution of Policy-Level Exogeneity Support for U.S. Budget Tax Measures

Note: The figure uses the budget-based U.S. tax measures with complete implementation dates, amounts, and policy-level structural probabilities that can be matched to the quarterly macro sample. The left panel plots the distribution of p, with vertical reference lines at 0.3, 0.5, and 0.7. The right panel reports the share of measures satisfying selected thresholds. In the current sample, there are no exact corner observations at $p = 0$ or $p = 1$.

4.3.4 From estimated support to quarterly tax shocks

Figure 5 translates the policy-level support measure into quarterly tax shocks. The left panel compares three series: all budget tax changes treated symmetrically, the hard-threshold budget shock that keeps measures with $p \geq 0.5$, and the original Romer–Romer exogenous tax shock series. The filtering step is economically consequential. It keeps 657 of the 1,078 matched budget measures and reduces total absolute shock mass from about 26.3 to 13.2 percent of GDP, indicating that the resulting series is not simply a rescaled version of the raw budget instrument.

The right panel makes this point directly. Relative to the unfiltered budget series, the hard-threshold construction retains less total shock mass while remaining denser than the RR benchmark. This is the intended role of the end-to-end procedure: preserve additional temporal support contributed by routine budget measures, but exclude the component of that support that receives weaker documentary backing for exogeneity.

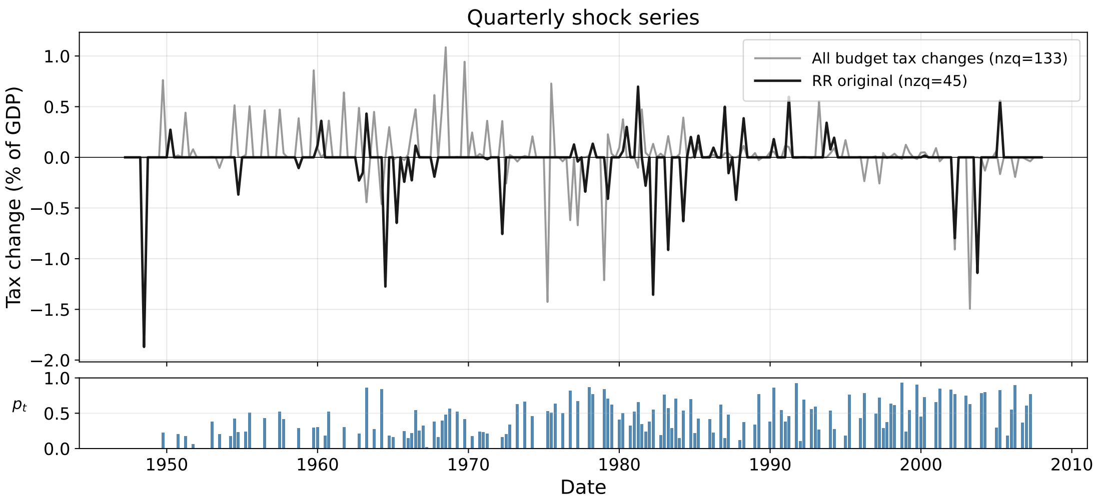

Figure 5: From Budget Measures to Quarterly Tax Shocks

Note: The left panel plots quarterly tax-shock series, all expressed as a percentage of contemporaneous GDP: all budget tax changes, the hard-threshold budget shock with $p \geq 0.5$, and the original Romer–Romer exogenous tax shock series. The right panel reports diagnostic aggregates for those same constructions. "nzq" denotes the number of non-zero quarters and $|s|$ denotes the sum of absolute quarterly shocks over the matched macro sample.

A useful diagnostic is that the budget-based shock series is not mechanically aligned with the original Romer–Romer exogenous series. Over the matched quarterly calendar, the correlation between the RR exogenous shock and the all-budget shock is -0.03. The correlation is also -0.03 for the hard-threshold budget shock with $p \geq 0.5$, and -0.02 for the hard-threshold budget shock with $p \geq 0.7$. The similarity of the macroeconomic responses below therefore does not come from reproducing the RR shock timing. It comes from selecting a distinct set of budget measures whose documentary support implies comparable identifying content.

4.3.5 Macroeconomic comparison with the Romer–Romer benchmark

For the end-to-end document exercise, the action-level evidence should use the same affine GMM estimator that is the baseline object in Section 1.4. The earlier local-projection comparison is useful as a bridge to the original Romer–Romer presentation, but it is not the same estimand as the support-indexed GMM response. I therefore report the full-quarter GMM endpoint for the action-level budget data alone. I do not include a separate RR-measure affine GMM analogue in the paper-facing endpoint figure, because that object is neither the published RR exogenous shock series nor the exact RR distributed-lag replication used above.

Figure 6 should therefore be read as the paper's action-level affine GMM endpoint, not as a direct RR replication exercise. The direct RR benchmark is the exact distributed-lag comparison reported earlier in this subsection; the GMM figure asks the narrower question of whether the document-based action universe produces a plausible high-support endpoint under the paper's baseline moment estimator.

The action-level GMM endpoint is contractionary over the medium run, with its largest decline around quarters 8–12. Appendix E.3 reports the corresponding low-support endpoint as a diagnostic for the affine specification.

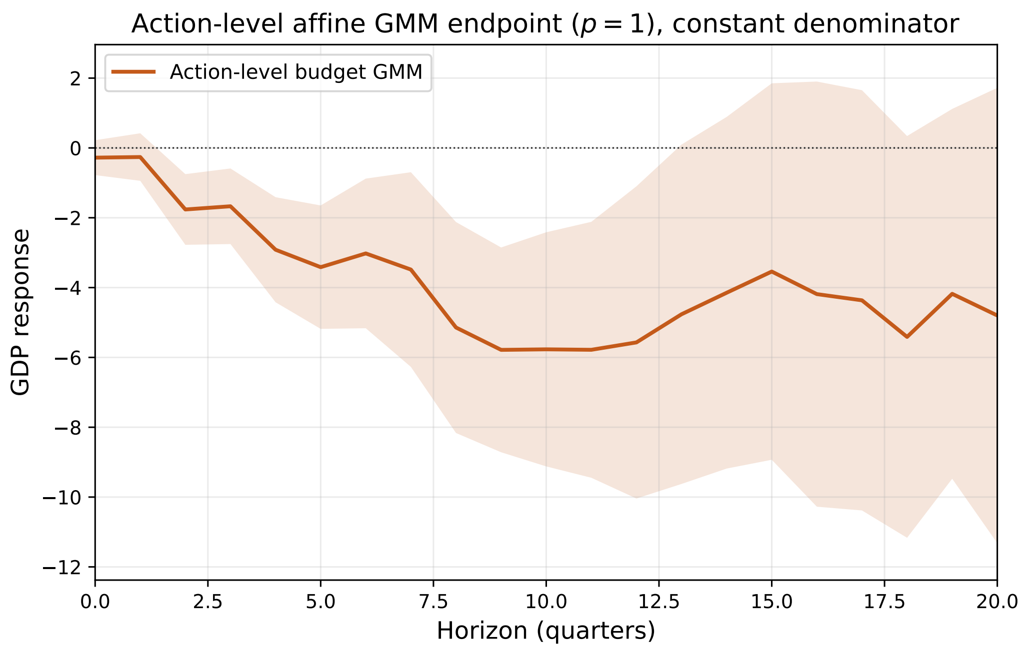

Figure 6: Action-Level Budget Shocks Under the Baseline GMM Estimator

Note: The figure reports the affine $p = 1$ endpoint response of log GDP estimated with the baseline full-quarter GMM estimator. The estimator residualizes GDP, inflation, and source-specific all-tax-shock controls, forms the numerator moment using $[D_t, D_t p_t]$, and reports $M_A/E[\tau_t^2 \mid D_t = 1]$ with 68% HAC confidence bands.

Taken together, the U.S. document exercise should be read as an out-of-sample extension of the RR identification logic. The exercise does not start from hand-written RR narratives and it does not attempt to reproduce the original episode list. Instead, it asks whether the document-grounded construction can recover economically meaningful identifying variation when applied to a denser universe of routine policy measures and their contemporaneous documentation. The answer from the baseline GMM environment is positive, but nuanced: the action-level endpoint is economically large, and its interpretation relies on the affine support-indexed moment rather than on a mechanical replication of the canonical RR series.

5 Conclusion

This paper revisits narrative identification in macroeconomics and argues that its core logic naturally admits a soft interpretation. Traditional narrative designs rely on binary inclusion rules that treat selected policy actions as fully exogenous while discarding the remainder. This paper generalizes that approach by allowing qualitative documentary evidence to enter continuously, reallocating identifying weight across policy actions according to the strength of narrative support for orthogonality in expectation. Identification is recast at the level of reduced-form moments, exploiting systematic variation in narrative support without requiring the recovery of observation-level structural policy shocks.

To operationalize this perspective, the paper develops a document-grounded implementation that translates policymakers' stated motivations into a continuous index of narrative support for exogeneity. The implementation systematizes the use of historical sources while preserving traceability to the underlying documentary record and incorporating diagnostics that discipline how textual evidence contributes to identification. These features are designed to support, rather than replace, economic judgment, and to make explicit the evidentiary basis underlying narrative restrictions.

The empirical analysis validates this object in stages. Edited Romer–Romer narratives show that the probability-construction stage assigns higher support to episodes classified as exogenous in the canonical U.S. narrative series. Cloyne's UK tax data then provide a full-pipeline external-label validation: probabilities constructed from reconstructed policy documents are systematically higher for tax changes that Cloyne classifies as exogenous. Finally, the end-to-end U.S. budget exercise applies the full procedure to routine federal tax measures outside the original Romer–Romer sample. Under the paper's baseline affine GMM estimator, the action-level budget endpoint generates a sizable contractionary GDP response, while the exact Romer–Romer distributed-lag figures remain the direct benchmark comparison.

More broadly, the framework clarifies how narrative evidence can discipline causal inference without requiring sharp exogeneity claims at the level of individual policy actions. By relaxing the implicit zero-contamination assumption underlying binary classifications, soft narrative identification retains information from a wider range of policy interventions while preserving the interpretive foundations of narrative approaches. This perspective helps address the sparsity of traditional narrative series and provides a principled way to trade off credibility and statistical power.

The analysis also illustrates how large collections of policy documents can be incorporated systematically into empirical macroeconomics when combined with transparent, document-grounded procedures. By lowering the cost of organizing and interpreting qualitative evidence, the framework expands the feasibility of narrative designs in settings characterized by frequent, lower-salience policy actions. Future work can build on this foundation by developing estimators that directly exploit variation in narrative support under weaker orthogonality assumptions. A natural substantive extension is to apply the same validated pipeline to new policy domains, including the French VAT application developed separately from this version of the paper.

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6 Appendix

A When a Two-Way Contamination Representation Is Valid

This appendix formalizes the scope of the two-way representation used in the main text. The contaminated component need not consist of homogeneous observations. It may pool several latent sources of bias, including sources with opposite signs. What matters for the validity of the two-way decomposition is not sign homogeneity, but whether the *average* contaminated moment remains stable conditional on the narrative support index.

Let S_t denote a latent state taking values in

$$\mathcal{S} = \{A, 1, \ldots, K\},$$

where $S_t = A$ denotes the orthogonal state and $S_t = k \in \{1, \ldots, K\}$ denotes one of K contaminated states. Let

$$p_t \equiv \Pr(S_t = A \mid \mathcal{D}_t)$$

be the narrative support index, where $\mathcal{D}_t$ is the contemporaneous documentary record. For each p in the support of p_t, define

$$\pi_k(p) \equiv \Pr(S_t = k \mid p_t = p), \qquad k = 1, \ldots, K,$$

so that

$$p + \sum_{k=1}^{K} \pi_k(p) = 1.$$

For each horizon $h \geq 0$, define the state-specific cross-moments

$$r_h^A \equiv \mathbb{E}(y_{t+h}\Delta\tau_t \mid S_t = A), \qquad r_h^k \equiv \mathbb{E}(y_{t+h}\Delta\tau_t \mid S_t = k).$$

The observable conditional outcome-tax moment is therefore

$$r_h(p) \equiv \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p) = p r_h^A + \sum_{k=1}^{K} \pi_k(p) r_h^k, \tag{10}$$

which allows the contaminated part to be arbitrarily heterogeneous. The question is when this richer structure can still be summarized by the simpler two-way representation used in the main text.

Proposition .1 (Validity of the two-way representation)**.** *There exists a constant r_h^B such that, for all p in the support of p_t,*

$$r_h(p) = p r_h^A + (1-p) r_h^B, \tag{11}$$

if and only if

$$\frac{\sum_{k=1}^{K} \pi_k(p) r_h^k}{1-p}$$

is constant in p on that support.

Proof. Starting from (10), define the average contaminated outcome-tax moment conditional on p by

$$\bar{r}_h^B(p) \equiv \frac{\sum_{k=1}^K \pi_k(p) r_h^k}{1-p}, \qquad p < 1.$$

Then

$$r_h(p) = p r_h^A + (1-p)\bar{r}_h^B(p).$$

Hence a representation of the form (11) with a constant r_h^B exists if and only if $\bar{r}_h^B(p)$ is constant in p. □

Proposition .1 shows that the two-way decomposition does not require all contaminated observations to have the same sign of bias or to be generated by a single latent source. It requires only that, conditional on the narrative support index, the contaminated observations can be summarized by a stable average moment. In this sense, the two-way representation is a parsimonious reduction of the first moments relevant for identification, not a claim of homogeneity of all endogenous tax changes. The next corollary specializes the result to the case in which the contaminated component combines sources of positive and negative bias.

Corollary .2 (Positive and negative contamination)**.** *Suppose there are two contaminated states, denoted $+$ and $-$, with outcome-tax moments r_h^+ and r_h^-. Let*

$$\omega(p) \equiv \Pr(S_t = + \mid S_t \neq A,\ p_t = p),$$

so that $1 - \omega(p) = \Pr(S_t = - \mid S_t \neq A,\ p_t = p)$. Then the average contaminated cross-moment conditional on p is

$$\bar{r}_h^B(p) = \omega(p) r_h^+ + [1-\omega(p)] r_h^-. \tag{12}$$

If $r_h^+ \neq r_h^-$, the two-way representation

$$r_h(p) = p r_h^A + (1-p) r_h^B$$

holds if and only if $\omega(p)$ is constant in p.

Proof. The result follows immediately from (12). If $r_h^+ \neq r_h^-$, the average contaminated moment $\bar{r}_h^B(p)$ is constant in p if and only if the composition weight $\omega(p)$ is constant in p. □

Corollary .2 makes precise the sense in which a changing composition of bias can invalidate the two-way model. Opposite-sign sources of contamination are not a problem by themselves. They become a problem only when their relative incidence varies systematically with p_t, so that the average contaminated moment is no longer stable. By contrast, if positive and negative contamination are balanced in a way that does not vary with p_t, then the two-way representation remains valid for the purpose of recovering the orthogonal moment.

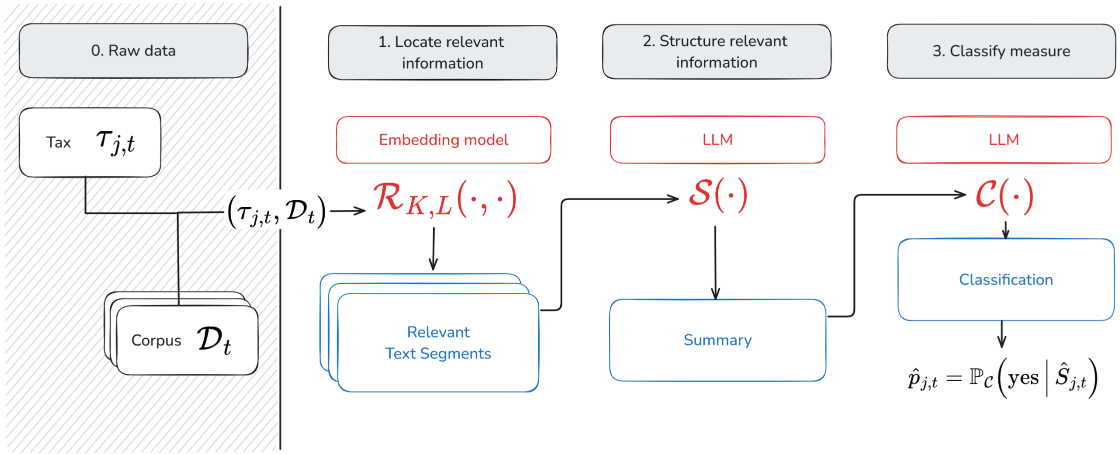

Figure 7: Mapping from policy descriptions and contemporaneous documents to narrative classifications. The figure illustrates the sequence $(q_i, \mathcal{D}_i) \to \hat{R}_i \to \hat{S}_i \to \hat{c}_i$ defined in Section 3.

B A Document-Grounded Implementation of Soft Narrative Identification

B.1 Illustration of the mapping

B.2 Embedding Model and Distance Metric

Semantic retrieval relies on vector representations of text segments and policy descriptions generated by a transformer-based embedding model. In the baseline implementation, embeddings are obtained using OpenAI's `text-embedding-3-large` model, which maps each text input to a 3,072-dimensional dense vector in $\mathbb{R}^{3072}$. The model is trained to capture semantic relationships between texts such that semantically similar passages are represented by vectors with small angular separation.

For each policy description q_i and document segment $d_{i,k} \in \mathcal{D}_i$, the corresponding embedding vectors e_i and $v_{i,k}$ are normalized to unit length. Semantic proximity is then evaluated using the cosine distance,

$$s_{i,k} = 1 - \frac{e_i \cdot v_{i,k}}{\|e_i\| \, \|v_{i,k}\|},$$

so that smaller values of $s_{i,k}$ indicate greater semantic similarity. Semantic proximity between text embeddings is measured using the cosine distance. Because all embeddings are normalized to unit length, cosine and Euclidean distances are mathematically equivalent in this setting, yielding identical retrieval rankings under either metric. Distances are computed through standard nearest-neighbor search routines provided by the underlying vector store, which efficiently identifies the text segments closest to each policy query in the embedding space. This normalization ensures that retrieval reflects semantic similarity rather than differences in vector magnitude. The retrieval operator defined in Equation (8) in the main text therefore selects, for each policy event i, the K document segments with the smallest cosine distances to the corresponding policy query q_i.

B.3 Example details

Applying the retrieval operator $\mathcal{R}_{K,L}$ to the query q_i and corpus $\mathcal{D}_i$, I obtain the top-K most relevant passages. Here are the top-5 passages:

- **Chunk 1.** On December 31, 1976, to mitigate, by a reduction of the weight of taxes bearing on consumption, the upward movement likely to occur at the end of this phase of price freeze; — it constitutes the last stage of a reform undertaken since 1969 — the date on which the standard rate of VAT was 23.456% — which results, ultimately, in a reduction, in less than seven years, of one quarter of the standard rate and in the merging of the latter with the intermediate rate.

- **Chunk 2.** An exceptional contribution would be requested in 1977 from the wealthiest French; it would affect all of them and only them, contrary to what had been provided for in the deleted article 5. This taxation would have as its base the total of the bases corresponding to the elements of lifestyle enumerated in article 168 of the General Tax Code, with the exception of main residences and cars of a power equal to or less than 16 horsepower. The income-equivalent threshold, calculated according to the norms of the aforementioned article 168, above which the exceptional contribution applies, is set at 60,000 F and the number of accumulated elements provoking taxation is three. Your Finance Committee proposes that you adopt the present article. VAT (Value Added Tax) – Article 6. Reduction of the standard rate and unification of the standard rate and the intermediate rate. Text. — I. — The standard rate of value-added tax is set at 17.60%. In the departments of Guadeloupe, Martinique and Réunion, it is set at 7.50II. — The upper limit of the special rebate for artisans fixed in paragraph 3 of article 282 of the General Tax Code is raised to 20,000 F. III. — The provisions of the present article enter into force on January 1, 1977. Comments. — It is proposed in the present article to lower by 2.4 points, starting from January 1, 1977, the standard rate of value-added tax which is currently 20% and, thereby, to bring it down to the level of the intermediate rate, namely 17.60%. This measure fits within a double perspective: — it is part of the program of combating inflation and is of a nature to permit, after the price freeze applied during the period from September 15 to December 31, 1976, to mitigate, by a reduction of the weight of taxes bearing on consumption, the upward movement likely to occur at the end of this phase of price freeze;

- **Chunk 3.** The number of rates applied in the matter of value-added tax, starting from next January 1, will be reduced to three instead of four; there would remain only: — the increased rate (33 %: cars, motorcycles, photography and cinematography equipment, sound and image recording devices, precious stones and precious metals, etc.); — the standard rate (17.6%: energy products, beverages, services and all transactions hitherto subject to the 20% rate, that is to say all those which were not taxed at the increased rate, nor at the intermediate rate, nor at the reduced rate); — the reduced rate (7%: essentially food products, pharmaceutical products, water and fertilizers). At the same time, it is planned to carry out a reduction of 2.5 points of the standard VAT rate and to align it with the intermediate rate in the departments of Guadeloupe, Martinique and Réunion, French Guiana not being concerned since the value-added tax regime is not applied there. In the three departments concerned, where, due to constraints inherent in the economies of developing countries, VAT rates are significantly lower than those of Metropolitan France, starting from January 1, 1977, one will record: — the increased rate: 14%; — the standard rate: 7.5%; — the reduced rate: 3.5%. In order to support the artisanal professions, it is proposed in paragraph II of the present article to raise, starting from January 1, 1977, the upper limit of the special rebate from which they benefit, raising it from 13,500 F to 20,000 F. Let us recall that, in order to lighten the burdens of small taxpayers, and in particular artisans, article 282 of the General Tax Code has provided, in their favor, special provisions resulting in a waiver or a reduction of the VAT normally incumbent upon them.

- **Chunk 4.** Standard of 6.5% per year. b) The elimination of the levy. As had been provided by the Finance Law for 1976, in the initial text presented by the Government, the levy was to cease to apply when the progression of prices was below a set norm: this one, in the governmental project, was fixed at 2.50% for the consumer price index of the group of manufactured products during a period of six consecutive months.

- **Chunk 5.** Article: — in paragraph I, the objective is to encourage business leaders to convert into capital the advances they are led to grant in the form of deposits in current accounts to the companies they manage. Up to now, in such cases, the ordinary contribution right was a proportional duty of 1%; it is planned to substitute for it a fixed duty of 220 F on the condition, however, that the sums thus incorporated into the capital have been placed at the constant disposal of the company, for a minimum period of twelve months.

B.4 Technical Implementation Details

The deterministic implementation relies on several technical specifications to ensure reproducibility. Both $\mathcal{S}$ and $\mathcal{C}$ operators use fixed prompt templates with the following decoding parameters: temperature = 0, top-p = 1.0, no nucleus sampling, and fixed maximum token limits. For transformer models, this eliminates stochastic sampling during generation.

For open-source models with frozen weights (e.g., LLaMA, Mistral family), deterministic decoding combined with identical computational environments guarantees bitwise-identical outputs across runs. This provides the strongest form of reproducibility, as the complete model architecture, weights, and inference code are publicly auditable.

For proprietary APIs (e.g., OpenAI GPT models, Google Gemini), additional logging is implemented to track potential sources of non-reproducibility. Each API call logs: (i) the specific model identifier and version, (ii) SHA-256 hashes of the prompt template and input content, (iii) all decoding parameters, (iv) timestamp and API response metadata, and (v) the complete generated output. This comprehensive logging enables detection of silent model updates or API changes that could affect results.

Even with deterministic settings, proprietary models may exhibit minor non-determinism due to distributed inference infrastructure, floating-point precision differences, or undocumented system changes. Additionally, model providers may silently update their systems, potentially altering outputs for identical inputs. While such changes are typically small, they can accumulate across large-scale classification tasks. The metadata logging system provides an audit trail that enables researchers to verify whether results were generated under identical conditions and to detect potential sources of variation in replication attempts.

C Grounding Diagnostics: Marginal Narrative Contributions

C.1 Robustness of the Linear Surrogate Estimator

Because the surrogate model is estimated with $n = 64$ ablations and up to $K \approx 30$ retrieved passages, I assess whether the resulting coefficients are sensitive to the choice of regularisation. Ridge regression yields estimates that are nearly identical to the OLS baseline and improves numerical stability when inclusion patterns exhibit collinearity. A LASSO with $\lambda = 0.01$ selects the same high-weight passages as OLS and produces coefficients of similar magnitude. These results indicate that the dependence patterns reported in the main text are not driven by estimator choice or overfitting. For settings with substantially larger K, penalised estimators would provide a natural extension.

C.2 Global Placebo Test for Summarisation Grounding

The coefficient-level placebo test in the main text evaluates, for each passage individually, whether its positive marginal contribution is unusually large relative to the distribution of positive placebo contributions. Because this diagnostic is applied to multiple passages within each event, some significant cells may be mechanically expected under the null of no grounding. This appendix reports an omnibus test that aggregates evidence across passages and verifies that the overall pattern of significant contributions cannot be attributed to chance.

Placebo mass. For each event, let $w_k^+ = \max\{\hat{w}_k, 0\}$ denote the positive marginal contribution of passage k, and let R and P denote the sets of relevant and placebo passages, respectively. The share of positive weight falling on placebo passages is

$$\phi = \frac{\sum_{k \in P} w_k^+}{\sum_{k \in R \cup P} w_k^+} \in [0, 1],$$

which summarises, at the event level, how much of the model's total positive dependence is placed on semantically irrelevant text. Grounded behaviour is characterised by low ϕ, while ungrounded dependence yields large values.

Randomisation inference. To assess whether the observed value ϕ_{obs} is unusually large, I test the sharp null that placebo labels are irrelevant to the pattern of marginal contributions. Let K^+ denote the number of passages with positive weights. Under the null, any assignment of K^+ placebo labels across passages is equally likely. I therefore compute the randomisation distribution by permuting placebo labels across the K^+ positions, recomputing placebo mass ϕ_b for each permutation, and forming the right-tailed p-value:

$$p = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{\phi_b \geq \phi_{\text{obs}}\}}{1 + B}.$$

A small p-value indicates that the model places significantly more positive weight on placebo passages than expected under random assignment, implying ungrounded behaviour. Conversely, when ϕ_{obs} falls well within the reference distribution, the observed coefficient-level significance pattern is unlikely to arise from multiple testing alone.

Interpretation. This global test complements the passage-level diagnostic in the main text. While the main procedure identifies *which* passages provide statistical support for each motivation, the global placebo mass test assesses *whether the overall pattern of support is itself meaningful.* Across events, the observed placebo masses cluster well below their randomisation-based critical values, confirming that the significant passages identified in the main text do not arise from chance alone.

D Application of the Document-Grounded Construction

D.1 Retrieval Performance Metrics and Parameter Selection [TO BE ADDED]

This Appendix provides additional detail on the evaluation of the retrieval operator $\mathcal{R}_{K,L}$, the calibration of its parameters (K, L), and the performance of alternative segmentation strategies. The objective is to document the robustness of retrieval accuracy and to clarify how the parameter settings used in the main text were chosen.

Benchmark Setup Each VAT measure i is manually linked to its corresponding Senate report article, denoted the *gold article*. For every candidate pair of parameters (K, L), I perform retrieval and record all text segments d_k whose span overlaps the gold article. These are counted as *hits*. Because each article can extend across multiple segments, a single event may have several true positives.

D.2 Classification Prompts and Category Scheme

This appendix reports the prompt templates used to classify value-added tax (VAT) measures into categories following the narrative taxonomy of Cloyne (2013). Each category corresponds to one of the subtypes summarized in Table 1. The classification operator $\mathcal{C}$ applies these prompts independently to the structured motivations $\hat{S}_i$ obtained from the summarization stage, returning binary (*Yes/No*) assessments and a textual rationale grounded in the retrieved evidence.

Prompt templates

Each prompt is presented in a standardized format and corresponds to one of the classification categories listed in Table 1. The prompts are organized by group—*endogenous* (Demand Management 1, Supply Stimulus 2, Deficit Reduction 3, Spending-driven 4) and *exogenous* (Long-term 5, Ideological 6, External 7, Deficit Consolidation 8). Each prompt provides the task definition, inclusion and exclusion criteria, and output format used in the classification procedure. Together, they operationalize the decision rules summarized in Table 1, ensuring full transparency and reproducibility of the classification process.

English translation of the classification prompt (Demand Management / DM):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Demand Management (DM)"?

Definition of Demand Management (DM):
A passage is classified as Demand Management if the measure is motivated by the need to influence, in the short term, the level or composition of aggregate demand in the economy—either to curb inflationary pressures and cool overheated sectors, or to stimulate demand during a slowdown or recession.

This includes:

- Tax increases or restrictions explicitly justified as reducing consumption, restraining demand, or mitigating inflationary pressures.
- Tax cuts or reliefs explicitly presented as stimulating demand, consumption, investment, or economic activity.
- Measures related to concerns over the balance of payments, inflation control, or unemployment, where the mechanism operates through aggregate demand.
- Temporary or reversible measures (e.g., surcharges, freezes, deferrals) justified by immediate economic circumstances, particularly short-term demand conditions.

*Do **not** classify as Demand Management if:*

- The motivation is solely deficit reduction or fiscal consolidation.
- The measure is driven exclusively by environmental, health, or equity objectives without reference to aggregate demand.
- The change is a routine indexation or neutral adjustment without intent to influence demand.
- The measure is presented as a long-term structural reform (improving incentives or competitiveness) rather than targeting short-term demand effects.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 1: Classification Prompt for the "Demand Management (DM)" Category

English translation of the classification prompt (Supply stimulus / SS):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Supply stimulus (SS)"?

Definition of Supply stimulus (SS):
A passage is classified as Supply stimulus if the measure is motivated by the need to stimulate, in the short term, production, investment, competitiveness, or employment on the supply side of the economy. Such measures typically involve tax cuts, reliefs, or structural changes aimed at improving incentives, profitability, liquidity, or business conditions to increase output and growth potential.

This includes:

- Reductions in income tax rates, allowances, or thresholds explicitly presented as improving work incentives or effort.
- Corporate or business tax cuts justified as encouraging investment, profitability, or competitiveness.
- Investment allowances or reductions in social contributions intended to lower hiring costs or promote employment.
- Measures described as supporting recovery or short-term growth through improvements on the supply side.
- Cyclical adjustments designed to maintain the immediate competitiveness of a sector.

Do ***not*** *classify as Supply stimulus if:*

- The justification emphasizes long-term structural, productivity, or competitiveness effects.
- The measure is presented as deficit reduction or fiscal consolidation.
- The measure targets aggregate demand management (short-term demand stimulation or restraint).
- The justification is primarily environmental, health-related, or social without a supply-side argument.
- The change is a neutral indexation, a temporary demand stimulus, or a routine adjustment without an incentive-based rationale.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 2: Classification Prompt for the "Supply stimulus (SS)" Category

English translation of the classification prompt (Deficit Reduction / DR):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Deficit Reduction (DR)"?

Definition of Deficit Reduction (DR):
A measure is classified as Deficit Reduction if its objective is to generate immediate revenue or prevent a loss of revenue in order to contain the short-term budget deficit.

This includes:

- The freezing or restriction of tax allowances or reliefs to raise funds for the national treasury "this year" or "next year."
- Increases in taxes or duties explicitly presented as necessary to raise additional revenue (e.g., "I need to raise X in additional receipts," "to finance this," "given the need to generate extra revenue").
- The elimination or reduction of costly reliefs whose expense is deemed unsustainable or unjustified.
- Adjustments to consumption taxes—such as fuel, alcohol, tobacco, or VAT—to "protect real yields" or prevent erosion of the tax base.
- Temporary or exceptional levies introduced to cover revenue needs within the current fiscal year.

*Do **not** classify as Deficit Reduction if:*

- The measure is primarily motivated by medium- or long-term fiscal consolidation (this belongs to another category).
- The measure is justified mainly by non-fiscal motives (environmental, health, or social equity) without clear reference to deficit control.
- The measure reflects routine indexation or neutral adjustments that do not raise revenue.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 3: Classification Prompt for the "Deficit Reduction (DR)" Category

English translation of the classification prompt (Spending-driven / SD):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Spending-driven (SD)"?

Definition of Spending-driven (SD):
A passage is classified as Spending-driven if the measure is explicitly motivated by the need to finance, offset, or enable increases in public spending or the introduction of new expenditure programs.

This includes:

- Tax increases or restrictions on reliefs introduced to cover the cost of new spending or the expansion of existing programs (e.g., food subsidies, NHS funding, social reforms).
- Statements indicating that additional revenue is needed to "finance," "help finance," or "enable" proposed spending measures.
- Exceptional or one-off taxes whose purpose is to fund a specific expenditure program (e.g., employment initiatives, child poverty measures).
- Reallocation or withdrawal of tax reliefs to free resources for targeted spending programs.

*Do **not** classify as Spending-driven if:*

- The motivation is deficit reduction or fiscal consolidation for long-term sustainability.
- The measure is justified by demand management (inflation control, balance-of-payments adjustment, or demand stimulus).
- The change is primarily motivated by environmental, health, or equity considerations without explicit reference to spending needs.
- The measure is a routine indexation or neutral adjustment with no link to expenditure financing.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 4: Classification Prompt for the "Spending-driven (SD)" Category

English translation of the classification prompt (Long-term / LR):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Long-term (LR)"?

Definition of Long-term (LR):
A passage is classified as Long-term if the measure is motivated by structural, supply-side, or long-term incentive effects on the economy—aimed at improving productivity, competitiveness, investment, saving, or economic efficiency over time.

This includes:

- Tax cuts or reforms justified by improving work effort, incentives, or morale.
- Measures designed to stimulate business investment, capital formation, or technological modernization.
- Reforms targeting international competitiveness, efficiency, or simplification of the tax system.
- Policies encouraging entrepreneurship, risk-taking, or saving to support future growth.
- Changes presented as necessary for the transition to a new economic regime (e.g., from wartime to peacetime).

Do ***not*** *classify as Long-term if:*

- The main motivation is short-term revenue needs or deficit consolidation.
- The measure concerns only short-term demand management (stimulus or demand restraint).
- The justification is purely environmental, health-related, or equity-driven without reference to long-term incentives or growth.
- The change is a routine indexation or minor administrative adjustment with no structural intent.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 5: Classification Prompt for the "Long-term (LR)" Category

English translation of the classification prompt (Ideological / IL):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Ideological (IL)"?

Definition of Ideological (IL):
A passage is classified as Ideological if the measure is primarily motivated by political values, social philosophy, principles of fairness, or a broader vision of society and the tax system—rather than by short-term macroeconomic management (demand control) or fiscal balance (deficit reduction or consolidation).

This includes:

- References to fairness, justice, tradition, or "social values," including environmental or ethical considerations.
- Tax reliefs or exemptions targeted at "deserving" groups (e.g., retirees, widows, children, families, vulnerable populations).
- Explicitly moral, social, or political justifications (e.g., "people hate income tax more than any other tax," "to reward work," "to encourage saving," "to support family unity").
- Long-term reforms aimed at making the tax system simpler, fairer, or more logical, without references to deficit or demand management.
- Measures presented as fulfilling electoral promises or party commitments (e.g., VAT cuts on essential goods, new redistributive taxes).
- Reforms introduced as part of a broader vision of taxation reflecting social or political values (e.g., taxing capital gains "as income," inheritance thresholds for family fairness).

Do ***not*** *classify as Ideological if:*

- The measure is primarily motivated by deficit reduction, fiscal sustainability, or budget consolidation.
- The change is framed in terms of short-term demand management (stimulus, inflation control, balance of payments).
- The measure is a routine indexation or a neutral technical adjustment.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 6: Classification Prompt for the "Ideological (IL)" Category

English translation of the classification prompt (External / EXT):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "External (EXT)"?

Definition of External (EXT):
A passage is classified as External if the measure is motivated by external economic pressures, trade obligations, or international agreements.

This includes:

- Measures adopted in response to balance-of-payments problems, exchange rate pressures, or the need to protect reserves.
- Tax modifications explicitly linked to supporting exports, foreign trade operations, or competitiveness in international markets.
- Adjustments mandated by commitments to international organizations or treaties (e.g., WTO, EU, EFTA).
- Policy changes justified by concerns about investor confidence or market reactions to the government's financing needs.

Do ***not*** *classify as External if:*

- The motivation is purely domestic (e.g., deficit reduction, consolidation, demand management, social equity, environment).
- The change is a routine adjustment unrelated to trade, finance, or international agreements.
- The measure concerns only domestic consumption, revenue, or distribution without an external framing.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 7: Classification Prompt for the "External (EXT)" Category

English translation of the classification prompt (Deficit Consolidation / DC):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Deficit Consolidation (DC)"?

Definition of Deficit Consolidation (DC):
A passage is classified as Deficit Consolidation if the measure is motivated by a structural or medium- to long-term strategy aimed at strengthening public finances or reducing deficits, borrowing, or public debt.

This includes:

- Measures presented as part of a sustained fiscal plan to ensure the soundness of public finances.
- Structural revenue generators (e.g., base broadening, duty escalators, restrictions on reliefs) intended to improve the budget balance over several years.
- Policies explicitly linked to fiscal consolidation, fiscal sustainability, or medium-term deficit reduction, even when other justifications are also mentioned.
- Measures whose initial intent was deficit consolidation, even if the rhetoric later evolved.

*Do **not** classify as Deficit Consolidation if:*

- The motivation concerns short-term revenue needs ("to raise additional revenue this year," "to finance support for the economy")—this falls under Deficit Reduction (DR), not Consolidation.
- The change is justified mainly by environmental, health, or social equity objectives without a fiscal framing.
- The measure reflects routine indexation or neutral adjustments without a consolidation goal.
- The policy reduces taxes or increases spending without compensating measures.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 8: Classification Prompt for the "Deficit Consolidation (DC)" Category

E Replication of Romer and Romer

E.1 Probability Benchmark Settings

E.1.1 Narrative extraction

This appendix documents the benchmark implementation behind Section 4.1. The edited RR narratives are used to validate the probability pipeline rather than a direct binary relabeling exercise. Each example below shows the portion removed before the narrative is passed to the summarization stage. As in the main text, the deleted block includes the explicit RR classification sentence and the two immediately preceding sentences. Removing these passages prevents the model from seeing the authors' bottom-line judgment while leaving enough narrative detail for the summarizer to recover multiple stated motivations from the remaining text.

Example 1: an endogenous measure

The motivation for the Revenue Act of 1951 was again the increase in spending related to the Korean War. The *1951 Economic Report* stated: "These new taxes are required to finance the defense effort; and to help keep total spending within the capacity of current production, so that inflation does not reduce the purchasing power" (p. 17). This same sentiment was echoed in a number of presidential speeches. On February 2, 1951, Truman said, "If we do not tax ourselves enough to pay for defense expenditures, the Government will ... add to total purchasing power and inflationary pressures" (*Special Message to the Congress Recommending a "Pay as We Go" Tax Program*, p. 2). These quotations make it clear that the tax increase was designed to keep growth normal.

It stated: "The military action in Korea, coupled with the general threat to world peace, has made it necessary to provide extraordinary increases in revenues to meet essential national defense expenditures" (82d Congress, 1st Session, House of Representatives Report No. 586, 6/18/51, p. 1). It also stressed the temporary nature of the tax, saying: "The $12\frac{1}{2}$-percent flat across-the-board increase [in individual income tax rates] was selected as the form of increase, since it is not intended that this increase will be permanent, and therefore, it was desired to provide an increase which would not become an integral part of the rate structure" (p. 8). Because the tax increase was designed to counteract the effects of higher military spending and thereby keep growth normal, I classify it as an endogenous, spending-driven action.

Truman's speech upon signing the bill stated that it would raise about \$5.5 billion of additional revenues at an annual rate (*Statement by the President Upon Signing the Revenue Act of 1951*, 10/20/51, p. 1). Both the *1952 Economic Report* (p. 135) and the *1951 Treasury Annual Report* (p. 44) gave the figure as \$5.4 billion. I follow my usual practice and use the number from the *Economic Report*.

The increase in corporate taxes was retroactive to April 1, 1951, while the remainder took effect on November 1, 1951 (*1951 Treasury Annual Report*, p. 51). Because the action took place before the middle of the quarter, I date the tax increase in 1951Q4. Approximately \$2.3 billion of the tax increase was from corporate taxes (p. 501). Therefore, the retroactive portion of the bill increased taxes at an annual rate of \$2.3 billion · 2 extra quarters, or \$4.6 billion. Adding this to the steady-state effect of \$5.4 billion yields a tax increase of \$10 billion in 1951Q4. The return to the steady-state level of tax increase implies a tax cut of \$4.6 billion in 1952Q1. If one ignored the retroactive part of the legislation, there would be just a tax increase of \$5.4 billion in 1951Q4.

The tax increase largely took the form of an increase in marginal rates. The act also raised the capital gains tax, the tax on corporate profits, and some excise taxes (*1951 Treasury Annual Report*, pp. 50–52).

Most of the changes were explicitly temporary. The individual income tax increases were legislated to expire on January 1, 1954; the corporate and excise tax increases were to continue until March 31, 1954 (*1951 Treasury Annual Report*, pp. 51–52).

Example 2: an exogenous measure

This bill was a significant change of the income tax system. It was the first comprehensive revision of the Internal Revenue Code in 75 years (*Statement by the President Upon Signing Bill Revising the Internal Revenue Code*, 8/16/54, p. 1).

This bill was passed primarily to increase the fairness and efficiency of the tax system. It had been under discussion since 1952 (Holmans, 1961, p. 219), well before the 1953–54 recession. At the signing ceremony, Eisenhower stressed: "It is a law which will help millions of Americans by giving them fairer tax treatment than they now receive." He also said, "economic growth will be fostered by such provisions as more flexible depreciation and better tax treatment of research and development costs, thus encouraging all business—large and small—to modernize and expand" (*Statement by the President Upon Signing Bill Revising the Internal Revenue Code*, 8/16/54, p. 1). The Report of the Committee on Ways and Means to Accompany H.R. 8300 (Internal Revenue Code of 1954) repeated this motivation. It stated: "In general, the purpose of these changes has been to remove inequities, to end harassment of the taxpayer and to reduce tax barriers to future expansion of production and employment" (83d Congress, 2d Session, House of Representatives Report No. 1337, 3/9/54, p. 1).

From this perspective, the reduction in revenues at a given level of income was the result of a belief that the bill would stimulate growth and of a desire to make the reform roughly revenue neutral after accounting for the growth effects. The *1954 Economic Report* stated, "A number of additional tax measures should now be taken in order to strengthen the forces of growth in employment and production. These measures, which involve some immediate sacrifices of revenue, contain the seeds of important future revenue gains to be reaped from the economic growth they will stimulate" (p. 79). The *1955 Economic Report* described the motivations for the bill as "to stimulate competitive enterprise, to strengthen the floor of security for the individual, and to curb tendencies toward either depression or inflation," and referred to "the loss of 1.4 billion dollars as a result of enacting structural tax changes" as almost a side effect of the reform (pp. 19–20).

The possible countercyclical benefits of the bill were certainly mentioned frequently. As discussed above, the administration favored some additional tax relief in mid-1954 because of further declines in defense spending. And, much of the Congressional debate focused on the need for additional stimulus. Indeed, several amendments were offered that involved increasing the personal exemption and cutting tax rates (Holmans, 1961, pp. 229–233). However, once the Excise Tax Reduction Act of 1954 was passed, the Administration argued strenuously against further tax cuts, especially the increase in the personal exemption.

[1] On March 15, 1954, Eisenhower said: "at this time economic conditions do not call for an emergency program that would justify larger Federal deficits and further inflation through large additional tax reductions" (*Radio and Television Address to the American People on the Tax Program*, 3/15/54, p. 4; see also *Annual Budget Message to the Congress: Fiscal Year 1955*, 1/21/54, p. 8). Also, the House report on the bill explicitly downplayed the countercyclical aspects, saying: "This bill is a long overdue reform measure which is vitally necessary regardless of the momentary economic conditions and should not be confused with other measures which may be, or might become, appropriate in the light of a particular

short-run situation" (House Report No. 1337, pp. 1–2). Thus, it seems clear that at least some of the tax cuts contained in the Excise Tax Reduction Act of 1954 and the Internal Revenue Code of 1954 should be classified as exogenous.

[2] Thus, it seems clear that at least some of the tax cuts contained in the Excise Tax Reduction Act of 1954 and the Internal Revenue Code of 1954 should be classified as exogenous. There was a cut beyond what the administration thought was justified by the decline in spending and the state of the economy. Because the excise tax reduction came first and had no obvious other motivation, I choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action.

[3] There was a cut beyond what the administration thought was justified by the decline in spending and the state of the economy. Because the excise tax reduction came first and had no obvious other motivation, we choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action. We feel this makes sense given that the Internal Revenue Code of 1954 clearly had a substantial exogenous component and indeed was passed after it was widely understood that the trough of the recession (May 30, 1954) had been turned (Holmans, 1961, p. 233).

[4] Because the excise tax reduction came first and had no obvious other motivation, we choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action. We feel this makes sense given that the Internal Revenue Code of 1954 clearly had a substantial exogenous component and indeed was passed after it was widely understood that the trough of the recession (May 30, 1954) had been turned (Holmans, 1961, p. 233). But, at some level, which of the two tax cuts is classified as endogenous and which as exogenous is somewhat arbitrary.

The act was signed in mid-August 1954, so I date the initial tax change in 1954Q3. My sources give a figure of –$1.4 billion for the revenue effects at an annual rate (for example, *1955 Economic Report*, p. 20; *1954 Treasury Annual Report*, p. 44). The bill was retroactive to January 1, 1954 (*1954 Treasury Annual Report*, p. 286). It thus reduced taxes by 3 · $1.4 billion, or $4.2 billion at an annual rate, in 1954Q3. The return to the steady-state reduction of $1.4 billion in 1954Q4 implies a tax increase of $2.8 billion in 1954Q4. If one chose to neglect the retroactive element, there would be just a tax cut of $1.4 billion in 1954Q3.

The changes consisted largely of the removal of obvious inequities, complications, ambiguities, and opportunities for tax avoidance, and of provisions designed to stimulate investment. All the changes were legislated to be permanent.

E.2 Romer and Romer replication Robustness checks

E.2.1 Probabilistic validation against the RR classification

As a complementary check, I estimate a simple probabilistic model for whether the original Romer–Romer label is exogenous. Let

$$R_i = \mathbf{1}\{\text{episode } i \text{ is classified as exogenous by Romer–Romer}\}.$$

I fit the one-variable calibration model

$$\Pr(R_i = 1 \mid p_i) = \Lambda\left(a + b \log \frac{p_i}{1 - p_i}\right),$$

where $\Lambda(\cdot)$ is the logistic cdf. The null $b = 0$ corresponds to no monotone predictive content in the policy-level probability. This exercise is deliberately exploratory: the RR classification is used as an external benchmark, not as a definitive latent truth, and the sample contains only 50 policy episodes.

Table 6: Exploratory Logit Validation Against the Original RR Classification

Statistic	**Estimate**
Number of RR episodes	50
Coefficient on $\log(p/(1-p))$	0.936
Standard error	0.331
Wald p-value for $b = 0$	0.005
Likelihood-ratio p-value	< 0.001
McFadden pseudo-R^2	0.181
In-sample AUC	0.761
Leave-one-out AUC	0.736

Note: The dependent variable equals one for episodes classified as exogenous in the original RR classification. p is the robust policy-level probability used in Table 3. The leave-one-out AUC refits the one-variable logit after dropping each episode in turn. The exercise is intended as a monotonicity and calibration check, not as a standalone classifier.

The positive and statistically significant slope indicates that higher policy-level probabilities are systematically associated with the original RR exogenous classification. At the same time, the AUC remains far below one. This is consistent with the interpretation in the main text: the probability contains meaningful narrative information, but ambiguous policy episodes are not perfectly separable by a single continuous score.

E.2.2 Probabilistic validation against the Cloyne classification

I apply the same one-variable probabilistic check to the Cloyne validation sample. Let

$$C_i = \mathbf{1}\{\text{policy row } i \text{ is classified as exogenous by Cloyne}\}.$$

I estimate

$$\Pr(C_i = 1 \mid p_i) = \Lambda\left(a + b \log \frac{p_i}{1 - p_i}\right),$$

where the sample consists of Cloyne policy rows with usable canonical labels and complete probability inputs. Because many policy rows belong to the same narrative event, I report both row-level and event-clustered uncertainty, and I also evaluate prediction after holding out entire narrative events.

Table 7: Exploratory Logit Validation Against the Cloyne Classification

Statistic	**Estimate**
Number of Cloyne policy rows	2,101
Coefficient on $\log(p/(1-p))$	0.551
Row-level standard error	0.054
Row-level Wald p-value for $b = 0$	< 0.001
Event-clustered standard error	0.100
Event-clustered Wald p-value for $b = 0$	< 0.001
Likelihood-ratio p-value	< 0.001
McFadden pseudo-R^2	0.042
In-sample AUC	0.629
Leave-one-event-out AUC	0.612
Event-grouped 5-fold AUC	0.621
Accuracy from raw $p \geq 0.5$	0.640

Note: The dependent variable equals one for policy rows classified as exogenous by Cloyne. Event-clustered inference clusters rows by narrative event. The leave-one-event-out AUC refits the one-variable logit after dropping each event group in turn; the event-grouped 5-fold AUC holds out folds of entire event groups. The exercise is intended as a monotonicity check against an external expert label, not as a standalone classifier.

The Cloyne probability is therefore statistically predictive of the external expert label: the slope is positive and remains precisely estimated after clustering by narrative event. The predictive strength is still modest, however. The AUC is about 0.63 in sample and around 0.61–0.62 under event-held-out evaluation. This is consistent with the main-text table: the probability ranks Cloyne-exogenous rows above Cloyne-endogenous rows on average, but the two distributions overlap substantially. I interpret this as evidence of monotone external-label validity, not as evidence that the procedure mechanically reproduces Cloyne's binary taxonomy.

E.2.3 Training-data leakage robustness check

To assess whether the language model has memorized specific source material, I implement a sentence-reconstruction test taken from Golchin and Surdeanu's (2024) contamination detection framework. The intuition is the following: if a model has encountered a document during training, it may be able to reproduce missing content from that document with unusual precision—especially when provided with cues pointing to the original source. To implement this test, I systematically remove sentences from the Romer and Romer narratives and asks the model to reconstruct them under two conditions: one relying solely on the surrounding context, and the other explicitly guiding the model by referencing the source paper and section. If the model performs significantly better under guided prompting, this may suggest prior exposure to the original text. To quantify and evaluate the reconstruction quality, I use both semantic (embedding cosine similarity) and lexical (ROUGE-L) metrics, followed by a bootstrap-based significance test to flag cases of potential memorization.

Using both metrics, and evaluating with `gpt-4o-mini` and `gemini-2.0-flash`, no narrative is flagged as a potential case of memorization. This result strengthens confidence that the model was not trained on Romer and Romer (2010), and supports the validity of the replication exercise by reducing concerns about training-data leakage as a source of bias in the model's classifications.

Procedure.

1. **Multiple draws & prompts.** For each narrative d I perform $T = 10$ draws. In draw t, one sentence $s_{d,t}$ is chosen uniformly at random, removed, and replaced by the token `[MISSING SENTENCE]`. I then issue two prompts a general prompt, and a guided prompt (Prompts 9).

> **General prompt:**
> You are provided with a passage in which one sentence has been removed and replaced with `[MISSING SENTENCE]`. Based ONLY on the context provided, generate a plausible sentence that could fill the gap. You do not need to match any specific source text — just ensure the result is coherent and contextually appropriate.
>
> {`narrative text from Romer and Romer`}
>
> Reconstructed sentence:

> **Guided prompt:**
> You are provided with a passage from the scientific paper *A Narrative Analysis of Postwar Tax Changes* by Christina D. Romer and David H. Romer, published in 2010. The passage comes from the section titled '{`title`}" in the main body of the paper.
> One sentence has been removed and replaced with `[MISSING SENTENCE]`. Based **only** on the context provided and your knowledge of this specific paper, reconstruct the missing sentence exactly as it appeared.
>
> {`narrative text from Romer and Romer`}
>
> Reconstructed sentence:

Prompt 9: General and Guided Prompt

The model returns two reconstructions $\hat{s}^{\text{gen}}_{d,t}$ and $\hat{s}^{\text{guide}}_{d,t}$.

2. **Similarity metrics.** For each original $s_{d,t}$ and each reconstruction, compute:

   - *Embedding cosine similarity* $\rho^{\text{gen}}_{d,t} = \cos\big(s_{d,t}, \hat{s}^{\text{gen}}_{d,t}\big)$, $\rho^{\text{guide}}_{d,t} = \cos\big(s_{d,t}, \hat{s}^{\text{guide}}_{d,t}\big)$, using `text-embedding-3-large`.
   - *ROUGE-L F1 score* $r^{\text{gen}}_{d,t}$ and $r^{\text{guide}}_{d,t}$ between the original and each reconstruction.

3. **Paired bootstrap test.** For each narrative d, define the mean differences

$$\Delta\rho_d = \frac{1}{T}\sum_{t=1}^{T}\big(\rho^{\text{guide}}_{d,t} - \rho^{\text{gen}}_{d,t}\big), \quad \Delta r_d = \frac{1}{T}\sum_{t=1}^{T}\big(r^{\text{guide}}_{d,t} - r^{\text{gen}}_{d,t}\big).$$

   We perform a nonparametric paired bootstrap with $B = 1{,}000$ resamples of the T pairs $\{(\rho^{\text{guide}}_{d,t}, \rho^{\text{gen}}_{d,t})\}$ (and similarly for $\{(r^{\text{guide}}_{d,t}, r^{\text{gen}}_{d,t})\}$), yielding bootstrap distributions of $\Delta\rho_d$ and Δr_d. Each indicator $\mathbf{1}\{\Delta\rho^{*(b)}_d \le 0\}$ (or $\mathbf{1}\{\Delta r^{*(b)}_d \le 0\}$) counts a bootstrap sample in which the similarity under the guided prompt is no better than—or worse than—that under the general prompt; that is, the model failed to improve reconstruction quality despite being explicitly cued about the source.

The empirical p-values are

$$p_d^\rho \;=\; \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}\{\Delta\rho_d^{*(b)} \le 0\}, \quad p_d^r \;=\; \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}\{\Delta r_d^{*(b)} \le 0\}.$$

Decision rule. A narrative d is flagged as potentially memorized if its bootstrap p-value for either similarity metric falls below 0.05, i.e. if guided prompting produces significantly closer reconstructions than general prompting.

E.2.4 Manually Investigating Legacy Binary Misclassifications

The misclassification pattern, mostly exogenous policies misclassified as endogenous, can possibly be attributed to an asymmetry inherent in macroeconomic narrative analysis: while it is relatively straightforward to construct narratives supporting the endogeneity of a policy intervention, establishing its exogeneity typically requires more stringent evidence. In the absence of sufficiently strong signals of such independence LLMs may default to a conservative interpretation, erring on the side of endogeneity.

Steering growth above normal levels. A recurring source of systematic misclassification involves tax cuts enacted with the aim of boosting growth above normal levels. In at least three instances, the LLM interprets these as countercyclical, and thus endogenous, due to the presence of growth-oriented motivation. However, according to Romer and Romer's decision rule, such motivations qualify as exogenous, as they are not designed to return output to trend but to exceed it. This kind of mislabeling can be mitigated by explicitly including this rule in the prompt design or few-shot examples, highlighting that aspirational growth policies, unless tied to negative output gaps, are to be treated as exogenous.

Conflicting motivations. Another set of misclassifications arises from cases with mixed or conflicting motivations. For example, one narrative discusses a tax increase intended primarily for deficit reduction and solvency of the Highway Trust Fund, both exogenous motives, but also references inflation concerns and contemporaneous spending increases. While the latter suggest endogenous intent, the original classification deems the measure exogenous with noted ambiguity. Such nuanced judgment is difficult for LLMs to replicate without additional guidance on prioritizing primary over secondary motivations.

"Package" related measure. Finally, there are highly specific edge cases that hinge on bespoke decision rules. A salient example is the Tax Adjustment Act of 1966. Despite being implemented to counteract high growth following an exogenous tax cut, Romer and Romer classify it as exogenous, arguing that treating it as endogenous would yield nonsensical results. This type of reasoning, context-specific and precedent-based, poses a significant challenge for LLMs, particularly in the absence of similar examples in the prompt.

E.3 Affine GMM Endpoint Diagnostic

The main text reports the high-support $p = 1$ endpoint of the baseline affine GMM estimator. Figure 8 reports the corresponding low-support $p = 0$ endpoint as a diagnostic. This contrast is useful for interpreting the affine specification: the contractionary medium-run response is concentrated at the high-support endpoint, while the low-support endpoint is much flatter and eventually positive. The $p = 0$ series should not be read as a separately observed class of purely endogenous policies, however, because the action-level probabilities are mostly interior and contain few observations close to the boundary. It is best interpreted as the low-support extrapolation implied by the affine moment model.

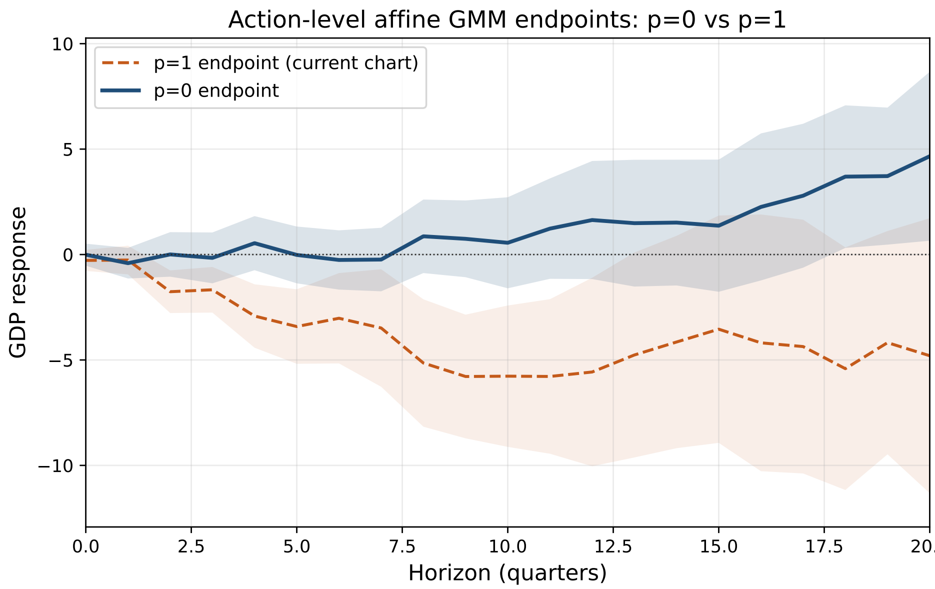

Figure 8: Affine GMM Endpoint Diagnostic: $p = 0$ and $p = 1$

Note: The figure reports the action-level U.S. budget affine GMM endpoints for real GDP. The solid blue curve is the implied $p = 0$ endpoint and the dashed orange curve is the paper-facing $p = 1$ endpoint. Both use the same full-quarter GMM estimator, residualization, event-constant denominator $E[\tau_t^2 \mid D_t = 1]$, and 68% HAC confidence bands. The $p = 0$ curve is a diagnostic extrapolation of the affine specification, not a separately observed policy class.

F Alternative LLM Signal: Incremental Log Probabilities

The main text uses hidden-state calibration as the baseline way to extract category probabilities from language models. A simpler alternative is to use the model's probability of a category-specific answer token. For motivation $M_{i\ell}$ and category $c \in \mathcal{G}$, let X_c be the category-specific prompt. Define the incremental log-probability signal

$$s^{\log}_{i\ell,c} := \log \Pr_{\text{LM}}(\text{Yes} \mid X_c \cup M_{i\ell}) - \log \Pr_{\text{LM}}(\text{Yes} \mid X_c). \tag{13}$$

The first term measures the model's support for category c after observing the motivation; the second removes the prompt-specific baseline tendency to answer "Yes." Collect these signals in

$$Z^{\log}_{i\ell} = \left(s^{\log}_{i\ell,c}\right)_{c \in \mathcal{G}}.$$

Because token-probability signals generated by different prompts need not be on a common probability scale, they should be treated as features to be calibrated rather than as final category probabilities. One can therefore estimate, on a calibration sample with known category labels, a multinomial mapping

$$\Pr(C_{i\ell} = c \mid Z_{i\ell}^{\log}) := \frac{\exp(\alpha_c + Z_{i\ell}^{\log\prime}\beta_c)}{\sum_{c' \in \mathcal{G}} \exp(\alpha_{c'} + Z_{i\ell}^{\log\prime}\beta_{c'})}, \qquad c \in \mathcal{G}. \tag{14}$$

The resulting fitted probabilities can then be aggregated exactly as in Section 2.2: sum the calibrated category probabilities over $\mathcal{G}_{\text{exo}}$ to obtain motivation-level support and average motivation-level support within the policy action to obtain p_i. This construction is useful as an alternative signal extraction method, while the baseline application uses the hidden-state calibrated probabilities reported in Table 1.

G Other

G.1 VAT table adoption

Table 8: VAT (or GST-equivalent) Adoption by Year in Developed Economies

Year	**Countries**
1954	France
1967	Denmark
1968	Germany
1969	Netherlands; Sweden
1970	Luxembourg; Norway
1971	Belgium
1972	Ireland
1973	Austria; Italy; United Kingdom
1977	South Korea
1980	Mexico
1985	Turkey
1986	New Zealand
1989	Japan
1990	Iceland
1991	Canada
1994	Finland
1995	Switzerland
2000	Australia